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Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**DESIGN CALCULATION** **3520 NEUTRALIZATION AND THICKENING THICKENER & TANK FOUNDATION**
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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management company(PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub-supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document

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1. INTENT

This document is to Prepare for design of **Process Water Neutralization and Thickening (3520)** structure's foundation. The analysis & design is carried out in accordance with project specifications,codes & standards. The design calculation ensures the safety and serviceability of the framing system.

2. SCOPE

In the specified area, there are 1 no of thickener i.e. THK-001 and 6nos reactor i.e. REA-001 to REA-006 & 1no underflow tank i.e. REA-007.
The analysis and design has been carried out in excel, considering the load data attached here with in Attachment 1. The load combinations for serviceability check and member's strength calculation are as per SBC.

3. DESIGN REQUIREMENTS

3.1. Documents & Codes

Documents (Refered):

i) Doc. No. MD-216-1000-EG-ST-CRT-1001 [Civil,Structural and Architectural Design Criteria]
ii) Doc. No. MD-101-SMEM-EG-ST-CRT-1001 [Structural Design Criteria - Process Building & Structures]
iii) Doc. No. MD-101-SMEM-EG-CV-CRT-1001 [Civil Design Criteria - General]

Codes (Refered) :

i) SBC 301 - 2007
ii) SBC 304-CR-18
iii) ACI 318
iv) ACI SP-17M(09)
v) API 650

3.2. Material Property

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

3.3. Grouting

- Grout shall be cementitious under steel column bases of Strength 40 MPa

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3.4. Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

3.5. Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to Sea Water or Soil	Foundation, Column & Tie Beam	75
Exposed to weather	Column & Floor Beam	50
	Wall and Slab	50

4. DESIGN CRITERIA

4.1. Design Loads

All Design Loads are indicated in Foundation Load Plan (Doc. No. MD-216-3520-EG-ME-PLN-1001/1002/1007 & MD-216-3520-EG-ME-GAD-1002) attached in Attachment 1. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

4.2. Load Combinations

For Neutralization Thickener
Load combinations are based on SBC - 301.
Basic Load Cases are –

1. Tank Dead Load & Bridge/Drive Dead Load(D)
2. Max. Fluid Load (F_{max})
3. Normal Operational Fluid Load ($F_{operating}$)
4. Seismic Load (E)
5. Wind Load (W)
6. Bridge/Drive Live Load(L)

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Allowable Stress Design

Load Case	Load Combination
101	$D+F_{max}$
102	$D+F_{max}+L$
103	$D+F_{max}+0.75L$
104	$D+F_{max}+W$
105	$D+F_{operating}+0.7E$
106	D+F+0.75L+0.75W
107	$D+F_{operation}+0.75L+0.75*(0.7E)$
108	0.6D+W
109	0.6D+0.7E

Ultimate Strength Design

Load Case	Load Combination
201	$1.4(D+F_{max})$
202	$1.4(D+F_{max})+1.7L$
203	1.2D+f1L
204	1.2D+0.8W
205	1.2D+1.6W +f1L
206	1.2D+1.0E +f1L
207	0.9D+1.6W
208	0.9D+1.0E

4.3. Stability against Overturning & Sliding

Overturning

Structures and buildings foundations, shall have the following minimum stability ratio against overturning:

- 1.5 For load combinations with wind or seismic loads.
- 2.0 For all other load combinations.

Sliding

A minimum factor of safety against sliding shall be 1.5.

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4.4. Allowable Soil Bearing Capacity

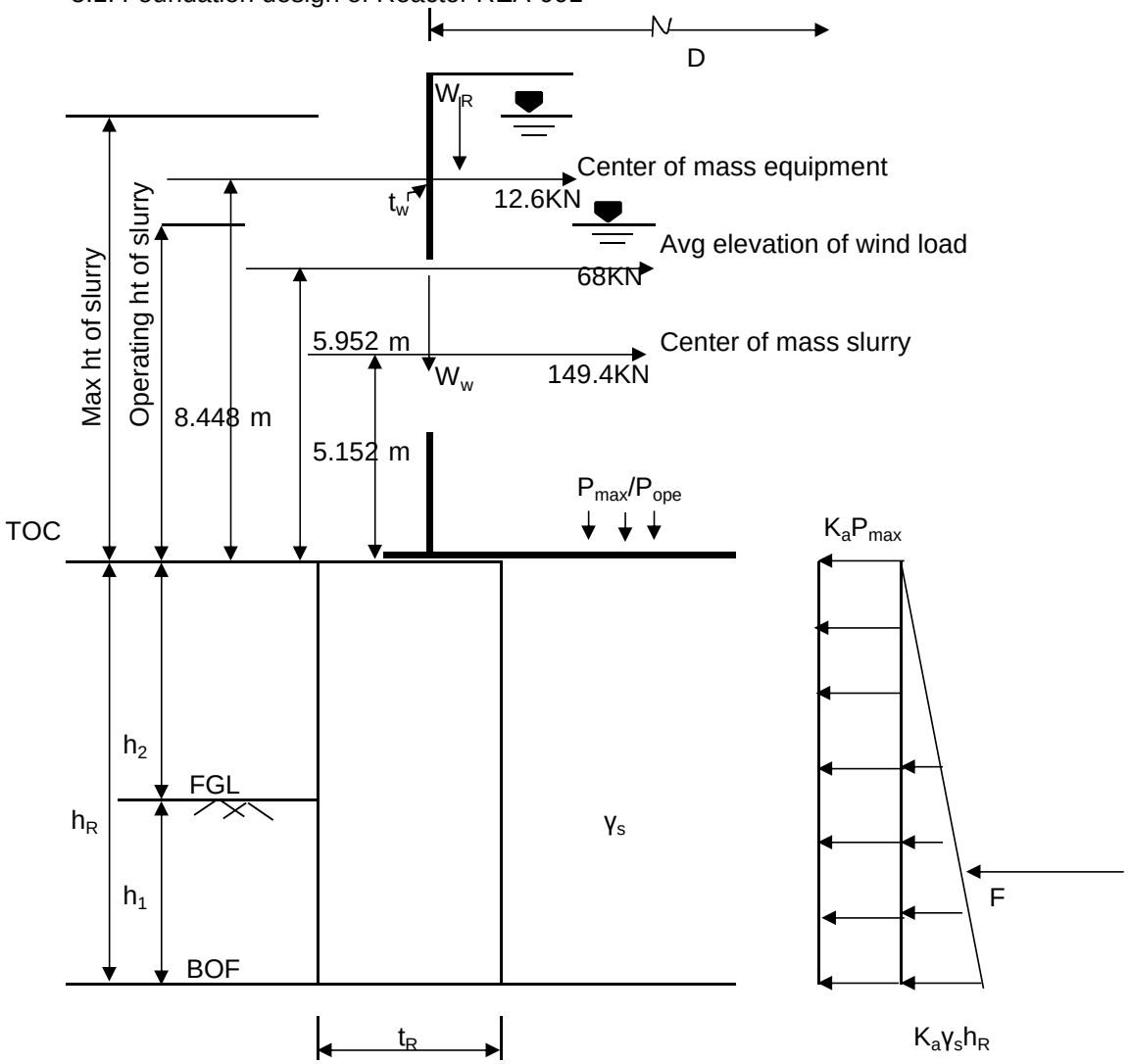
The net safe bearing capacity of soil for foundations design has been considered as 46 T/m^2 at 1.2m depth below EGL (RL 911.30m) as per L&T soil report MD-216-3000-EG-CV-RPT-1008-A01, bore hole BH-13.

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5. DESIGN CALCULATION OF REINFORCEMENT

5.1. Foundation design of Reactor REA-001



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	6.51	m
Self weight of tank =	299	KN
Self weight of tank per m of circumference,W_w =	14.6	KN/m
External platform weight =	31	KN
Self weight of tank roof per m of circumference,W_R=	1.52	KN/m
Maximum weight of slury=	3847	KN
Pressure at base of tank ,P_{max}	116	KN/m^2
Operating weight of slury=	3556	KN
Pressure at base of tank ,P_{ope}	107	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	450	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	3.15	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	1.95	m
Ringwall thickness, t_R =	400	mm
Inner dia of ringwall,D_i =	6.11	m
Outer dia of ringwall,D_o =	6.91	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	12.59	m^3

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 68 kn acting at a height of 9.102 m from BOF

M_{OT} = Overturning moment = 618.9 KN-m

M_{OT} / S_{xx} = 49.2 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 174 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 223 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 125 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 115.9 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 165.1 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 66.78 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 618.9 KN-m

M_{OR} = Restoring moment = (299+PI()*6.51*0.4*3.15*25)*6.51/2 = 3070.2 KN-m

FOS = M_{OR} / M_{OT} 4.96 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Seismic force at center of mass equipment = 299/(299+3556)*162 = 12.6 KN
acting at a height of 11.598 m from BOF

Seismic force at center of mass slurry = 3556/(299+3556)*162 = 149.4 KN
acting at a height of 8.302 m from BOF

M_{OT} = Overturning moment = 1386 KN-m

M_{OT} / S_{xx} = 110 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 169 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 246 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 92.3 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 173.7 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 175.4 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 151 Kn/m

Hoop tension in ringwall, T = 1/2FD = 492 Kn

Factored design tension, T_u= 689 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 1823 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 2520 mm^2

$A_{s,reqd}$ (per face of ring wall)= 1260 mm^2

Required : 12 dia bar @ 283 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 1782 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 480 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 240 mm^2

Required : 10 dia bar @ 327 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

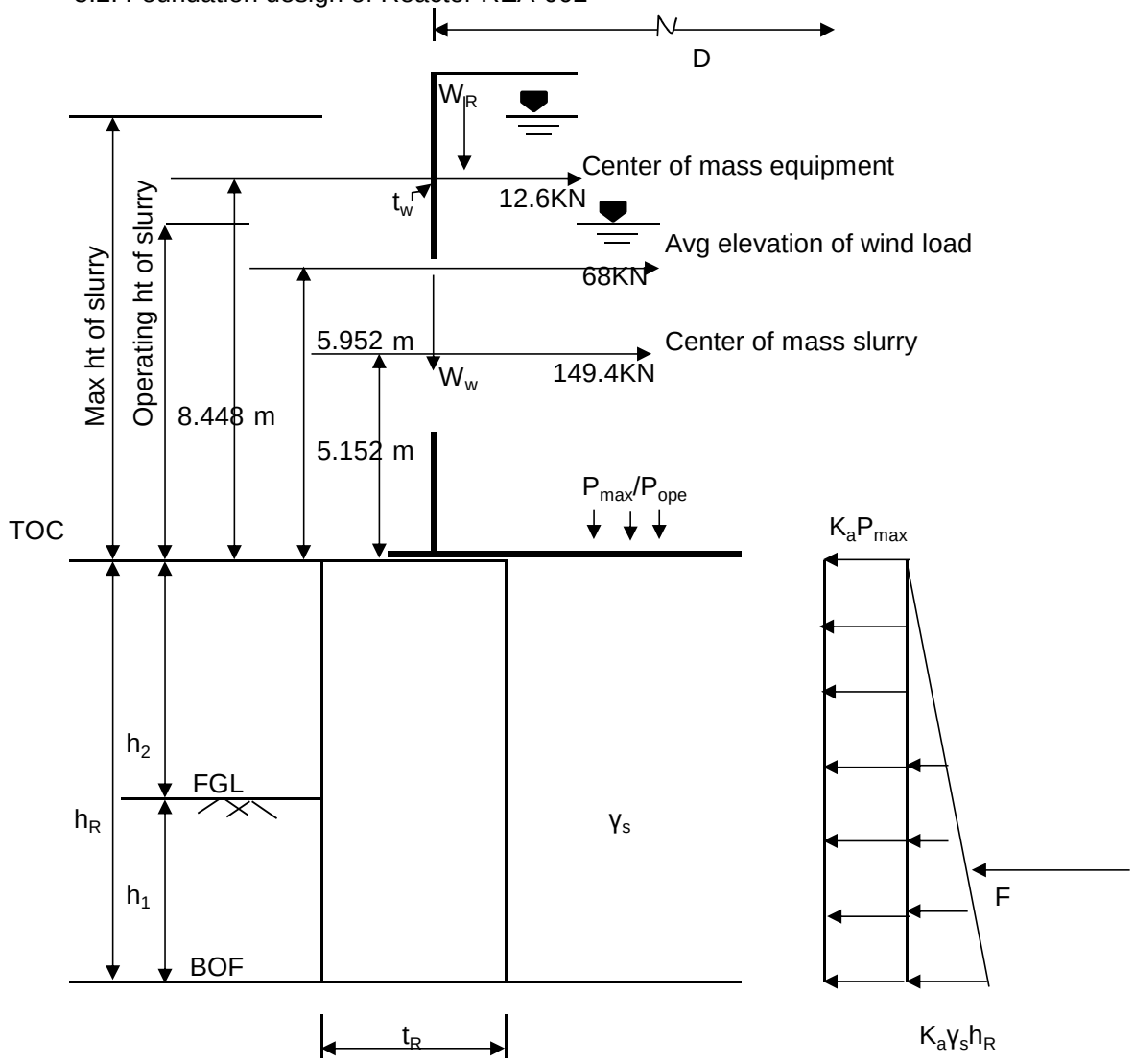
$A_{s,\ provided}$(per face) = 393 mm^2 ok

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5.2. Foundation design of Reactor REA-002



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	6.51	m
Self weight of tank =	299	KN
Self weight of tank per m of circumference,W_w =	14.62	KN/m
External platform weight =	31	KN
Self weight of tank roof per m of circumference,W_R=	1.516	KN/m
Maximum weight of slury=	3847	KN
Pressure at base of tank ,P_{max}	115.6	KN/m^2
Operating weight of slury=	3556	KN
Pressure at base of tank ,P_{ope}	106.8	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	450	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.94	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	1.74	m
Ringwall thickness, t_R =	400	mm
Inner dia of ringwall,D_i =	6.11	m
Outer dia of ringwall,D_o =	6.91	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	12.59	m^3

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 68 kn acting at a height of 8.892 m from BOF

M_{OT} = Overturning moment = 605 KN-m

M_{OT} / S_{xx} = 48 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 169 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 217 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 121 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 110.9 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 158.9 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 62.87 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 604.7 KN-m

M_{OR} = Restoring moment = (299+PI()*6.51*0.4*2.94*25)*6.51/2 = 2930.4 KN-m

FOS = M_{OR} / M_{OT} 4.85 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Seismic force at center of mass equipment = 299/(299+3556)*162 = 12.6 KN

acting at a height of 11.388 m from BOF

Seismic force at center of mass slurry = 3556/(299+3556)*162 = 149.4 KN

acting at a height of 8.092 m from BOF

M_{OT} = Overturning moment = 1352 KN-m

M_{OT} / S_{xx} = 107 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 164 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 240 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 89.1 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 168.7 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 171.4 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 139.2 Kn/m

Hoop tension in ringwall, T = 1/2FD = 453 Kn

Factored design tension, T_u= 634 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 1678 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 2352 mm^2

$A_{s,reqd}$ (per face of ring wall)= 1176 mm^2

Required : 12 dia bar @ 282.9 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 1663 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 480 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 240 mm^2

Required : 10 dia bar @ 327.4 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

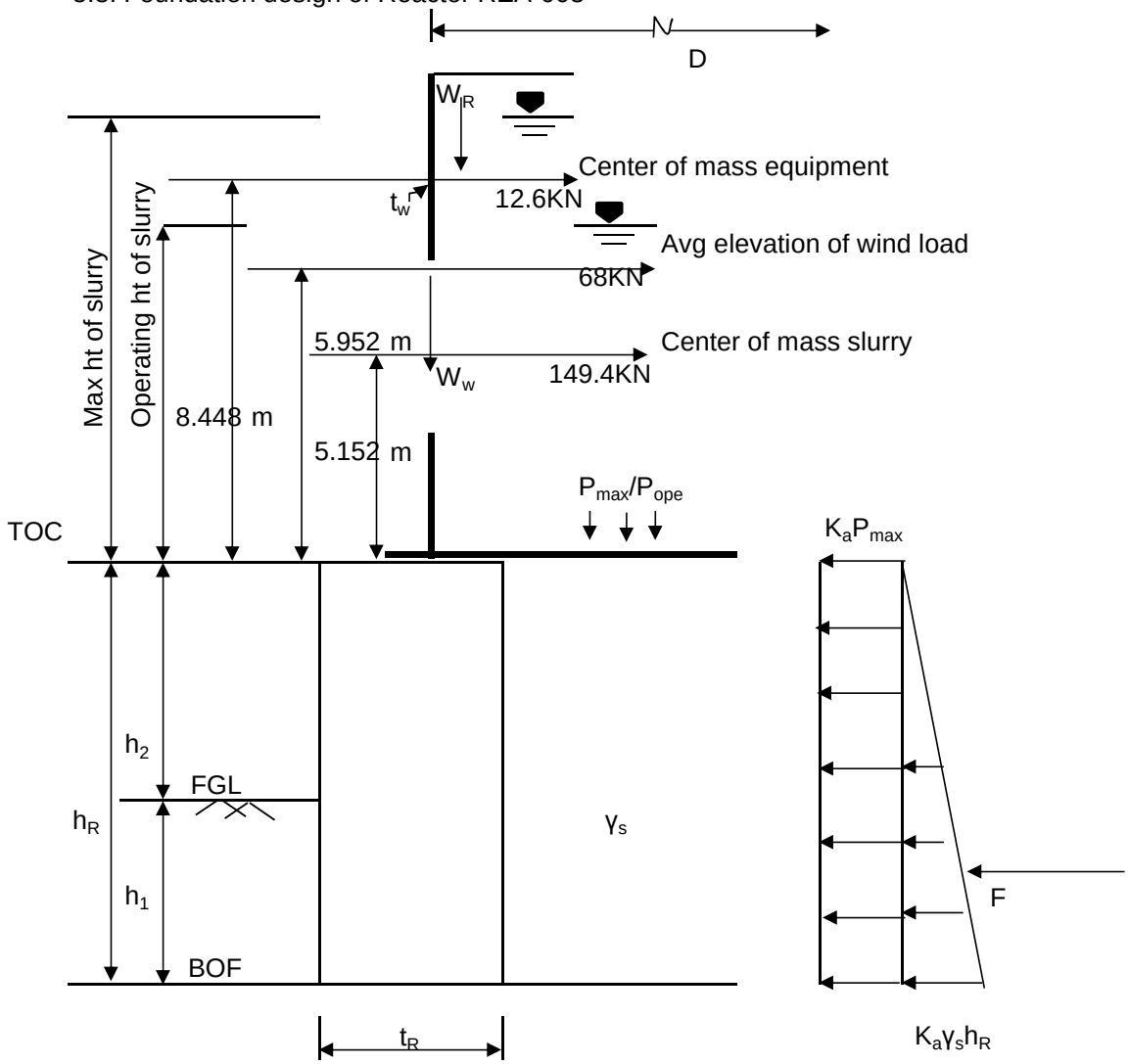
$A_{s,\ provided}$(per face) = 392.9 mm^2 ok

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5.3. Foundation design of Reactor REA-003



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	6.51	m
Self weight of tank =	299	KN
Self weight of tank per m of circumference,W_w =	14.6	KN/m
External platform weight =	31	KN
Self weight of tank roof per m of circumference,W_R=	1.52	KN/m
Maximum weight of slury=	3847	KN
Pressure at base of tank ,P_{max}	116	KN/m^2
Operating weight of slury=	3556	KN
Pressure at base of tank ,P_{ope}	107	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	450	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.73	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	1.53	m
Ringwall thickness, t_R =	400	mm
Inner dia of ringwall,D_i =	6.11	m
Outer dia of ringwall,D_o =	6.91	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	12.59	m^3

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 68 kn acting at a height of 8.682 m from BOF

M_{OT} = Overturning moment = 590 KN-m

M_{OT} / S_{xx} = 46.9 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 164 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 211 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 117 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 105.9 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 152.7 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 58.97 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 590.4 KN-m

M_{OR} = Restoring moment = (299+PI()*6.51*0.4*2.73*25)*6.51/2 = 2790.6 KN-m

FOS = M_{OR} / M_{OT} 4.73 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Seismic force at center of mass equipment = 299/(299+3556)*162 = 12.6 KN
acting at a height of 11.178 m from BOF

Seismic force at center of mass slurry = 3556/(299+3556)*162 = 149.4 KN
acting at a height of 7.882 m from BOF

M_{OT} = Overturning moment = 1318 KN-m

M_{OT} / S_{xx} = 105 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 159 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 233 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 86 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 163.6 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 167.4 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 127 Kn/m

Hoop tension in ringwall, T = 1/2FD = 415 Kn

Factored design tension, T_u= 581 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 1537 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 2184 mm^2

$A_{s,reqd}$ (per face of ring wall)= 1092 mm^2

Required : 12 dia bar @ 283 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 1544 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 480 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 240 mm^2

Required : 10 dia bar @ 327 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

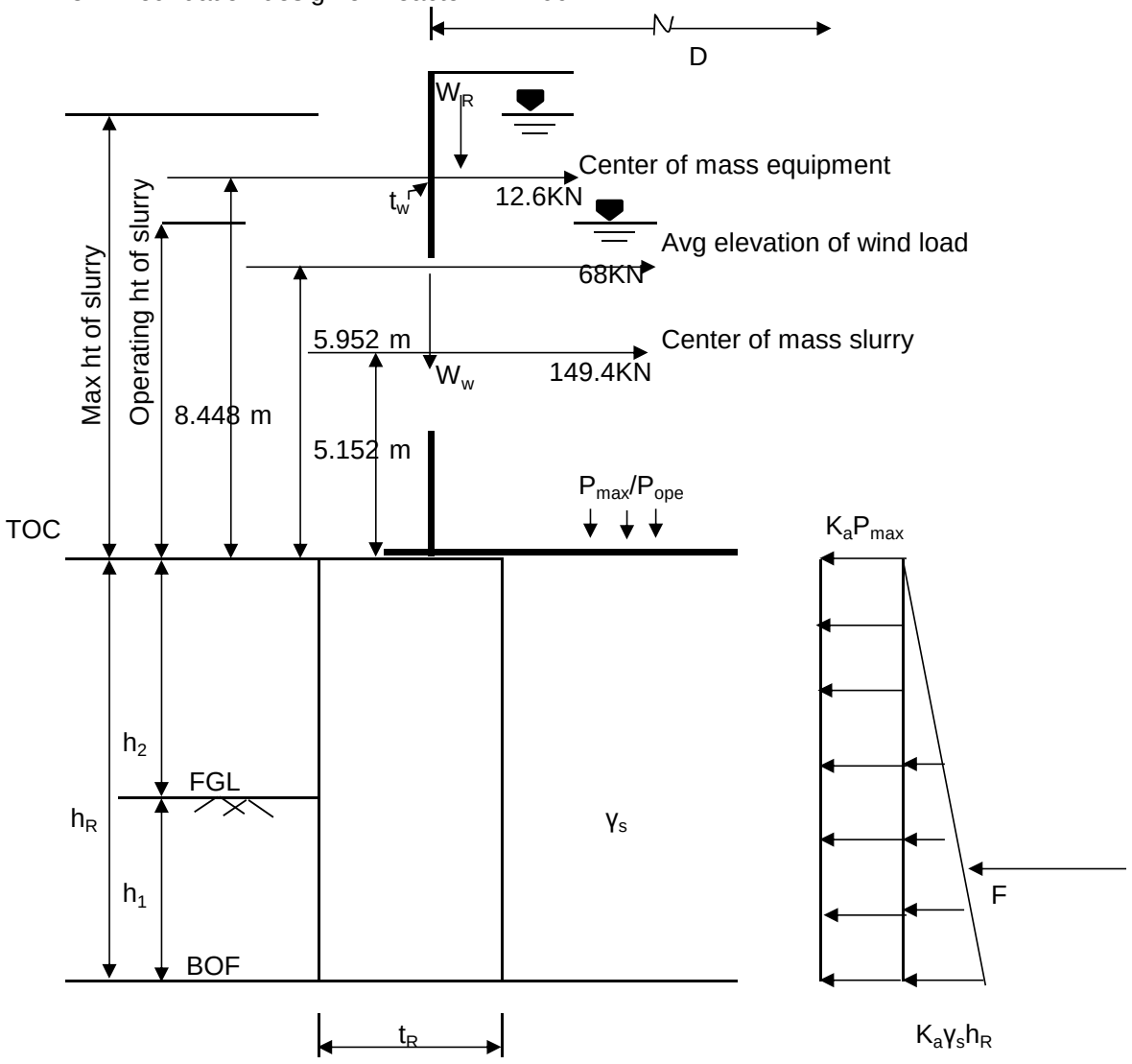
$A_{s,\ provided}$(per face) = 393 mm^2 ok

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5.4. Foundation design of Reactor REA-004



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	6.51	m
Self weight of tank =	299	KN
Self weight of tank per m of circumference,W_w =	14.62	KN/m
External platform weight =	31	KN
Self weight of tank roof per m of circumference,W_R=	1.516	KN/m
Maximum weight of slury=	3847	KN
Pressure at base of tank ,P_{max}	115.6	KN/m^2
Operating weight of slury=	3556	KN
Pressure at base of tank ,P_{ope}	106.8	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	450	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.52	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	1.32	m
Ringwall thickness, t_R =	400	mm
Inner dia of ringwall,D_i =	6.11	m
Outer dia of ringwall,D_o =	6.91	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	12.59	m^3

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 68 kn acting at a height of 8.472 m from BOF

M_{OT} = Overturning moment = 576 KN-m

M_{OT} / S_{xx} = 45.8 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 159 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 204 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 113 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 100.8 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 146.6 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 55.06 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 576.1 KN-m

M_{OR} = Restoring moment = (299+PI()*6.51*0.4*2.52*25)*6.51/2 = 2650.8 KN-m

FOS = M_{OR} / M_{OT} 4.60 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Seismic force at center of mass equipment = 299/(299+3556)*162 = 12.6 KN
acting at a height of 10.968 m from BOF

Seismic force at center of mass slurry = 3556/(299+3556)*162 = 149.4 KN
acting at a height of 7.672 m from BOF

M_{OT} = Overturning moment = 1284 KN-m

M_{OT} / S_{xx} = 102 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 154 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 226 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 82.8 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 158.6 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 163.5 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 116 Kn/m

Hoop tension in ringwall, T = 1/2FD = 378 Kn

Factored design tension, T_u = 529 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 1399 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 2016 mm^2

$A_{s,reqd}$ (per face of ring wall)= 1008 mm^2

Required : 12 dia bar @ 282.9 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 1426 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 480 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 240 mm^2

Required : 10 dia bar @ 327.4 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

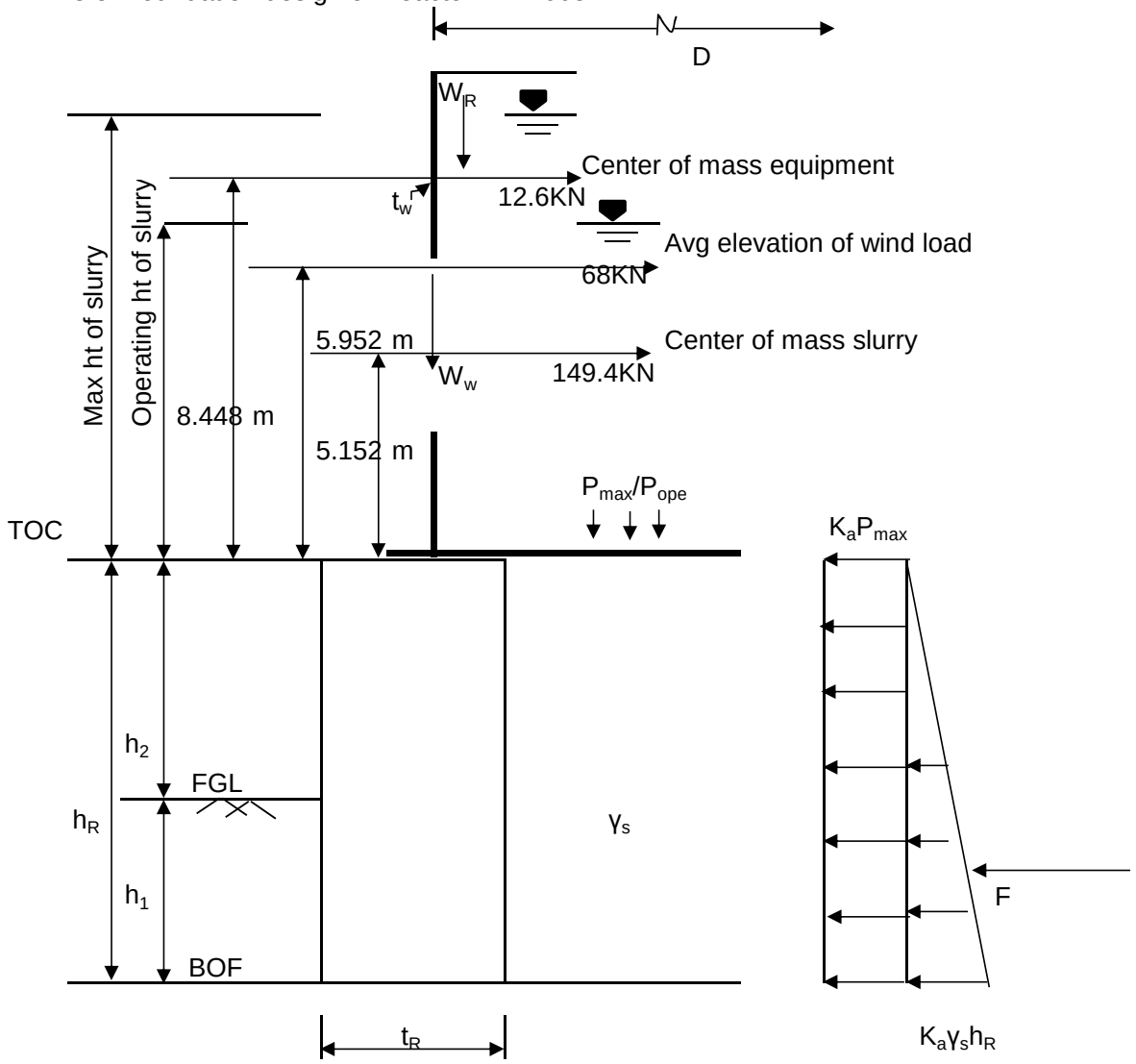
$A_{s,\ provided}$(per face) = 392.9 mm^2 ok

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5.5. Foundation design of Reactor REA-005



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	6.51	m
Self weight of tank =	299	KN
Self weight of tank per m of circumference,W_w =	14.62	KN/m
External platform weight =	31	KN
Self weight of tank roof per m of circumference,W_R=	1.516	KN/m
Maximum weight of slury=	3847	KN
Pressure at base of tank ,P_{max}	115.6	KN/m^2
Operating weight of slury=	3556	KN
Pressure at base of tank ,P_{ope}	106.8	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	450	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.31	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	1.11	m
Ringwall thickness, t_R =	400	mm
Inner dia of ringwall,D_i =	6.11	m
Outer dia of ringwall,D_o =	6.91	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	12.59	m^3

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 68 kn acting at a height of 8.262 m from BOF

M_{OT} = Overturning moment = 562 KN-m

M_{OT} / S_{xx} = 44.6 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 154 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 198 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 109 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 95.8 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 140.4 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 51.16 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 561.8 KN-m

M_{OR} = Restoring moment = (299+PI()*6.51*0.4*2.31*25)*6.51/2 = 2511.0 KN-m

FOS = M_{OR} / M_{OT} 4.47 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Seismic force at center of mass equipment = 299/(299+3556)*162 = 12.6 KN
acting at a height of 10.758 m from BOF

Seismic force at center of mass slurry = 3556/(299+3556)*162 = 149.4 KN
acting at a height of 7.462 m from BOF

M_{OT} = Overturning moment = 1250 KN-m

M_{OT} / S_{xx} = 99.3 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 149 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 219 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 79.7 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 153.6 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 159.5 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 104.8 Kn/m

Hoop tension in ringwall, T = 1/2FD = 341 Kn

Factored design tension, T_u= 478 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 1264 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1848 mm^2

$A_{s,reqd}$ (per face of ring wall)= 924 mm^2

Required : 12 dia bar @ 282.9 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 1307 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 480 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 240 mm^2

Required : 10 dia bar @ 327.4 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

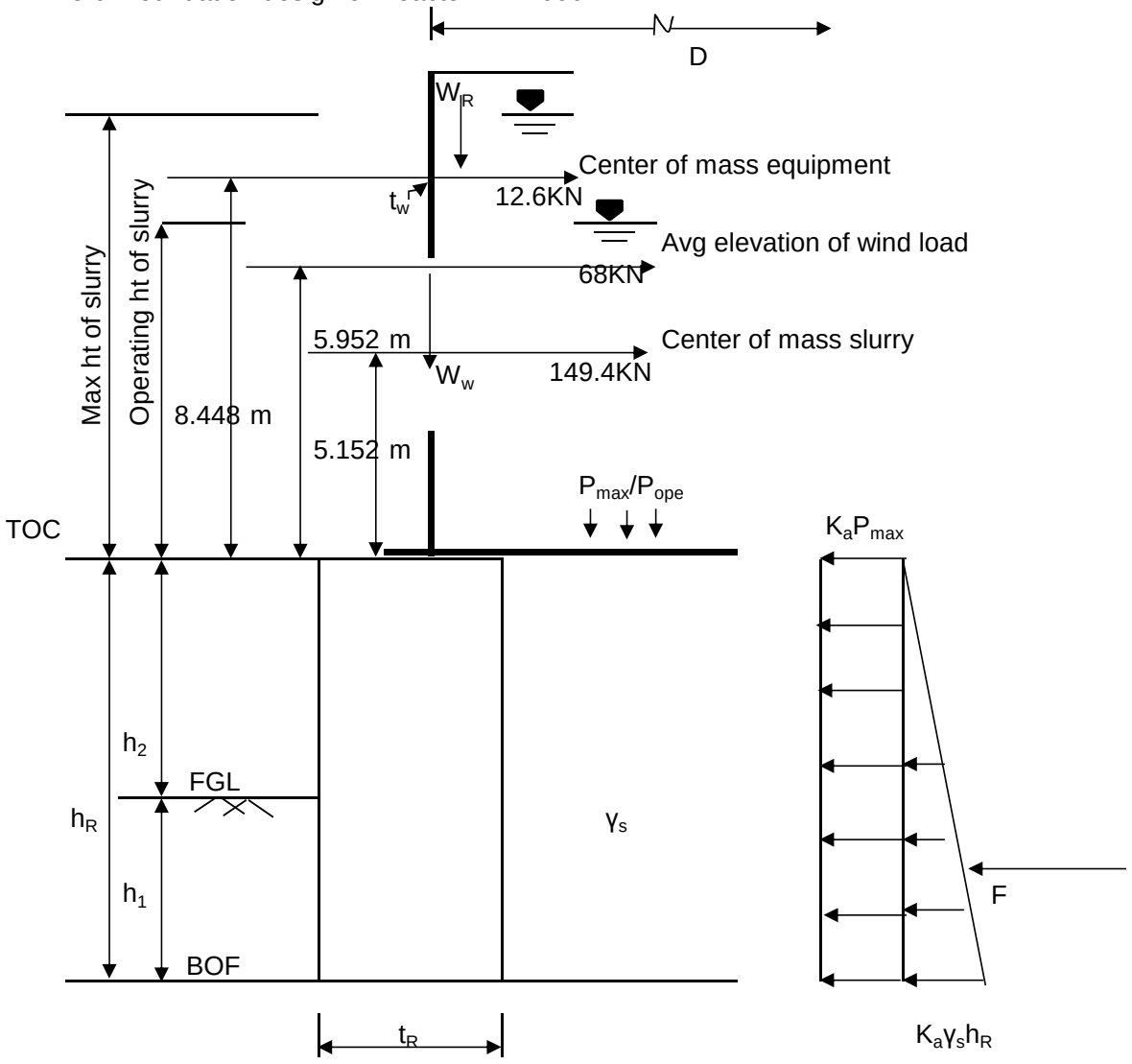
$A_{s,\ provided}$(per face) = 392.9 mm^2 ok

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5.6. Foundation design of Reactor REA-006



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	6.51	m
Self weight of tank =	299	KN
Self weight of tank per m of circumference,W_w =	14.62	KN/m
External platform weight =	31	KN
Self weight of tank roof per m of circumference,W_R=	1.516	KN/m
Maximum weight of slury=	3847	KN
Pressure at base of tank ,P_{max}	115.6	KN/m^2
Operating weight of slury=	3556	KN
Pressure at base of tank ,P_{ope}	106.8	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	450	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.1	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	0.9	m
Ringwall thickness, t_R =	400	mm
Inner dia of ringwall,D_i =	6.11	m
Outer dia of ringwall,D_o =	6.91	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	12.59	m^3

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 68 kn acting at a height of 8.052 m from BOF

M_{OT} = Overturning moment = 548 KN-m

M_{OT} / S_{xx} = 43.5 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 149 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 192 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 105 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 90.7 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 134.2 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 47.25 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 547.5 KN-m

M_{OR} = Restoring moment = (299+PI()*6.51*0.4*2.1*25)*6.51/2 = 2371.2 KN-m

FOS = M_{OR} / M_{OT} 4.33 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Seismic force at center of mass equipment = 299/(299+3556)*162 = 12.6 KN
acting at a height of 10.548 m from BOF

Seismic force at center of mass slurry = 3556/(299+3556)*162 = 149.4 KN
acting at a height of 7.252 m from BOF

M_{OT} = Overturning moment = 1216 KN-m

M_{OT} / S_{xx} = 96.6 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 144 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 212 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 76.5 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 148.5 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 155.5 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 93.92 Kn/m

Hoop tension in ringwall, T = 1/2FD = 306 Kn

Factored design tension, T_u= 428 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 1132 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1680 mm^2

$A_{s,reqd}$ (per face of ring wall)= 840 mm^2

Required : 12 dia bar @ 282.9 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 1188 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 480 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 240 mm^2

Required : 10 dia bar @ 327.4 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

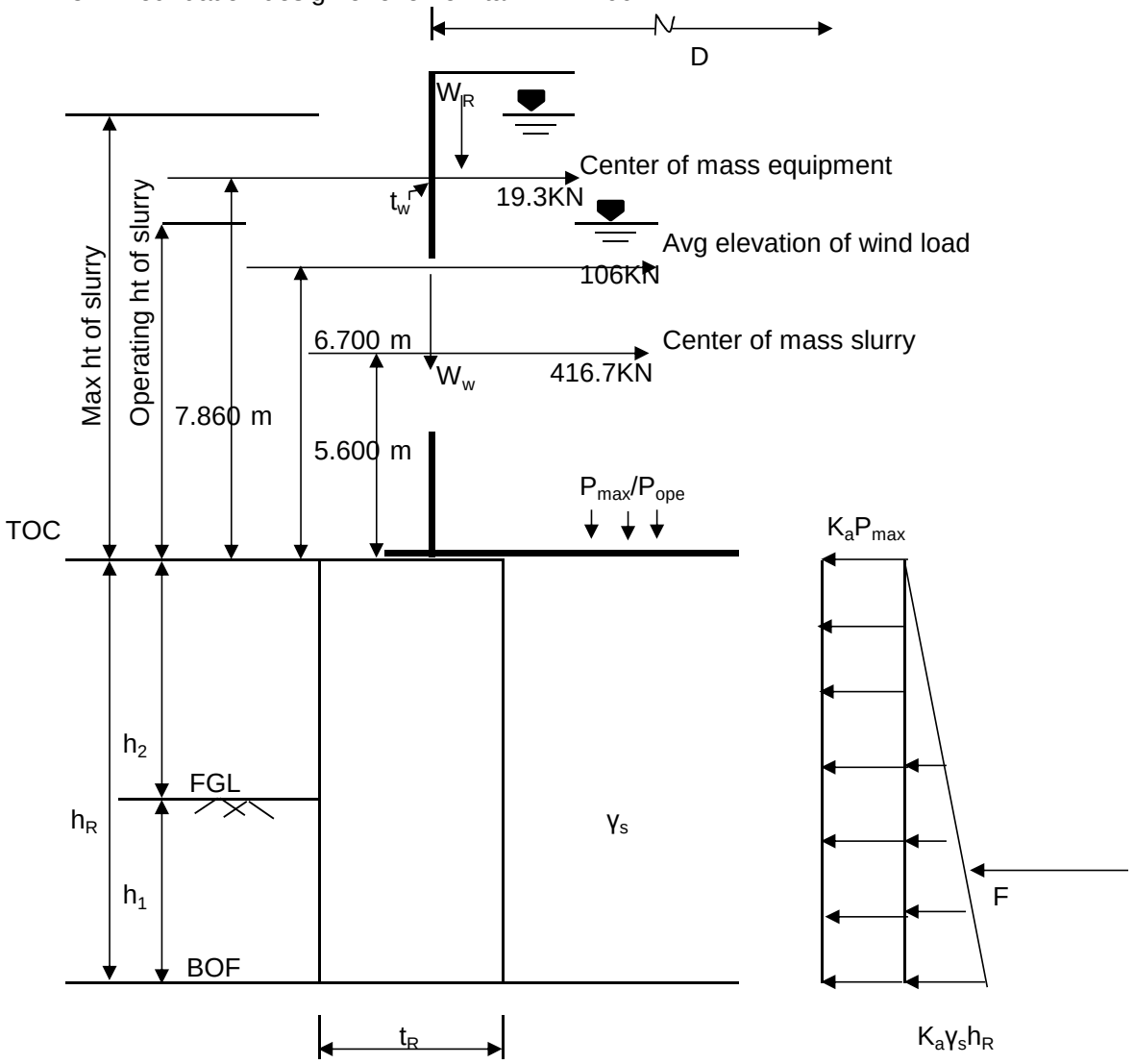
$A_{s,\ provided}$(per face) = 392.9 mm^2 ok

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5.7. Foundation design of overflow tank REA-007



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	10.012	m
Self weight of tank =	441.5	KN
Self weight of tank per m of circumference,W_w =	14.04	KN/m
External platform weight =	43.9	KN
Self weight of tank roof per m of circumference,W_R=	1.396	KN/m
Maximum weight of slury=	10669	KN
Pressure at base of tank ,P_{max}	135.5	KN/m^2
Operating weight of slury=	9527	KN
Pressure at base of tank ,P_{ope}	121	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	19	Kn/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	450	Kn/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	2.1	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	0.9	m
Ringwall thickness, t_R =	400	mm
Inner dia of ringwall,D_i =	9.612	m
Outer dia of ringwall,D_o =	10.41	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	30.33	m^3

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Check of soil bearing pressure (Max Slurry Load): For wind Load

Wind force = 106 kn acting at a height of 8.800 m from BOF

M_{OT} = Overturning moment = 933 KN-m

M_{OT} / S_{xx} = 30.8 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 157 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 187 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 126 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 89.0 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 119.7 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 58.23 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 932.8 KN-m

M_{OR} = Restoring moment = (441.5+PI()*10.012*0.4*2.1*25)*10.012/2 = 5516.7 KN-m

FOS = M_{OR} / M_{OT} 5.91 > 1.5 ok

Check for soil bearing pressure (Operating Slurry Load): For Seismic Load

Seismic force at center of mass equipment = 441.5/(441.5+9527)*436 = 19.3 KN

acting at a height of 9.960 m from BOF

Seismic force at center of mass slurry = 9527/(441.5+9527)*436 = 416.7 KN

acting at a height of 7.700 m from BOF

M_{OT} = Overturning moment = 3401 KN-m

M_{OT} / S_{xx} = 112 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = (W_w + W_R)/t_R + P_{ope}/2 + h_R\gamma_c$ = 149 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 228 KN/m^2 < 450.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 71 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = (W_w + W_R)/t_R + P_{max}/2 + h_R\gamma_c$ = 156.7 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 175.4 Kn/m^2 < 450.0 KN/m^2 Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 107.7 Kn/m

Hoop tension in ringwall, T = 1/2FD = 539 Kn

Factored design tension, T_u= 755 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 1998 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1680 mm^2

$A_{s,reqd}$ (per face of ring wall)= 998.8 mm^2

Required : 12 dia bar @ 237.9 c/c (per face)

Provide : 12 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 1188 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 480 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 240 mm^2

Required : 10 dia bar @ 327.4 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 392.9 mm^2 ok

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5.8. Foundation design of thickener THK-001

- Inner Column
- Load Calculation :

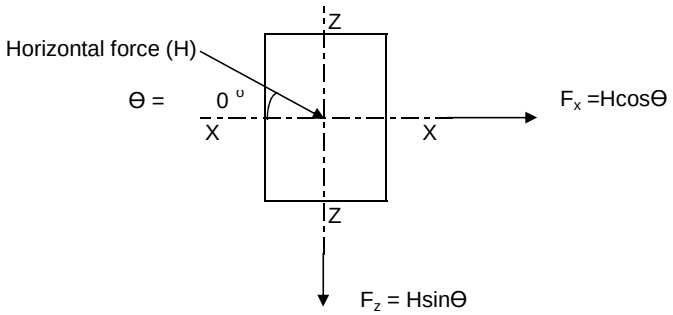
Input load data for Neurtalization thickener

Load Table :

	Tank dead load	Max fluid load		Normal operating fluid load		Seismic load			Wind load			Bridge/drive dead load	Bridge/drive live load
Central Column	kN	kN		kN		kN	kN	kN-m	kN	kN	kN-m	kN	kN
	V	H	V	H	V	H	V	Moment	H	V	Moment	V	V
	40	0	1910	0	1758	8	0	7	7	0	7	0	0

H : Horizontal load V : Vertical load

Horizontal reactions are in the plane of bracing.



Depth of foundation from U/S of base plate = 2.1 m

Load at bottom of foundation level

Primary load cases		F_x	F_y	F_z	M_x	M_z
		kN	kN	kN	kN-m	kN-m
Tank dead load	D	0	40	0	0	0
Selfweight of column = 0.6*0.6*(1.6-0.5)*25 =		0	9.9	0	0	0
Selfweight of footing =		0	66.125	0	0	0
Weight of soil above footing =		0	62	0	0	0
Bridge/drive dead load		0	0	0	0	0
Max fluid load	F_{max}	0	1910	0	0	0
Normal operating fluid load	$F_{operati}$	0	1758	0	0	0
Seismic load	E	8	0	0	0	23.8
Wind load	W	7	0	0	0	21.7
Bridge/drive live load	L	0	0	0	0	0

Combinations including seismic load to use normal operating fluid load.

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Load combinations for foundation servicibility check :

Sl no	Basic combinations as per cl 2.4.1 SBC 301	F_x	F_y	F_z	M_x	M_y	M_z
		kN	kN	kN	kN-m	kN-m	kN-m
101	$D+F_{max}$	0	2088	0	0	0	0
102	$D+F_{max}+L$	0	2088	0	0	0	0
103	$D+F_{max}+0.75L$	0	2088	0	0	0	0
104	$D+F_{max}+W$	7	2088	0	0	0	21.7
105	$D+F_{operating}+0.7E$	5.6	1936	0	0	0	16.7
106	D+F+0.75L+0.75W	5.25	2088	0	0	0	16.3
107	$D+F_{operation}+0.75L+0.75*(0.7E)$	4.2	1936	0	0	0	12.5
108	0.6D+W	7	132	0	0	0	21.7
109	0.6D+0.7E	5.6	132	0	0	0	16.7

Weight of soil above footing includes in all load combination except (0.6D+W or 0.6D+E).

Load combinations for strength check :

Sl no	Basic combinations as per cl 2.3.2 SBC 301	F_x	F_y	F_z	M_x	M_y	M_z	q_{net1}	q_{net2}	q_{net3}	q_{net4}	q_{avg}
		kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	kN/m^2	kN/m^2	kN/m^2	kN/m^2
201	$1.4(D+F_{max})$	0	2744	0	0	0	0	518.7	519	519	519	519
202	$1.4(D+F_{max})+1.7L$	0	2744	0	0	0	0	518.7	519	519	519	519
203	1.2D+f1L	0	60	0	0	0	0	11.32	11.3	11.3	11.3	11.3
204	1.2D+0.8W	5.6	60	0	0	0	17.36	19.88	2.76	19.9	2.76	19.9
205	1.2D+1.6W +f1L	11.2	60	0	0	0	34.72	28.44	-5.8	28.4	-5.8	28.4
206	1.2D+1.0E +f1L	8	60	0	0	0	23.8	23.06	-0.42	23.1	-0.42	23.1
207	0.9D+1.6W	11.2	44.91	0	0	0	34.72	25.61	-8.63	25.6	-8.63	25.6
208	0.9D+1.0E	8	44.91	0	0	0	23.8	20.23	-3.25	20.2	-3.25	20.2

f_1 = 1 cl 2.4.1 SBC 301

Design Parameter for net base pressure calculation:

Area of Footing , A = 5.29 m^2

Zz-z = 2.03 m^3

Zx-x = 2.03 m^3

Net Pressure, $q_{net} = P/A \pm M_z/Z_{zz} \pm M_x/Z_{xx}$

Consider average net pressure for design.

q_{avg} for design = 518.7 kN/m^2

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• Design of Column C1 :

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 2

Column Dimensions :

Depth parallel to X-X axis, h = 600 mm

Depth parallel to Z-Z axis, b = 600 mm

Clear Cover = 75 mm

d' = 95 mm

Y_h = (h-d')/h = 0.68

Y_b = (b-d')/b = 0.68

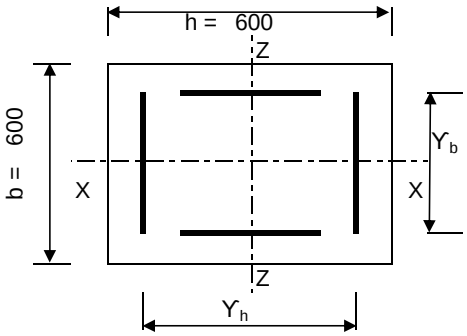
Unsupported length of column parallel to X-X axis, l_{ux} = 1.3 m kl_{ux}/b = 4.333 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.3 m kl_{uz}/h = 4.333

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 9746 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	kN	kN-m		kN	kN-m				kN-m			kN-m				
201	2743.9	0.0	0.0	4221	0.0	0.0	0.335	0.120	907.2	0.00	0.120	907.2	0.00	1.61	0	safe
202	2743.9	0.0	0.0	4221	0.0	0.0	0.335	0.110	831.6	0.00	0.110	831.6	0.00		0	safe
203	59.9	0.0	0.0	92.1	0.0	0.0	0.007	0.050	378	0.00	0.050	378	0.00		0	safe
204	59.9	0.0	17.4	92.1	0.0	26.7	0.007	0.050	378	0.00	0.050	378	0.07		0.02	safe
205	59.9	0.0	34.7	92.1	0.0	53.4	0.007	0.050	378	0.00	0.050	378	0.14		0.04	safe
206	59.9	0.0	23.8	92.1	0.0	36.6	0.007	0.050	378	0.00	0.050	378	0.10		0.02	safe
207	44.9	0.0	34.7	69.1	0.0	53.4	0.005	0.050	378	0.00	0.050	378	0.14		0.04	safe
208	44.9	0.0	23.8	69.1	0.0	36.6	0.005	0.050	378	0.00	0.050	378	0.10		0.02	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{lie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3600	12	20	+	0	0	3772	1.05	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

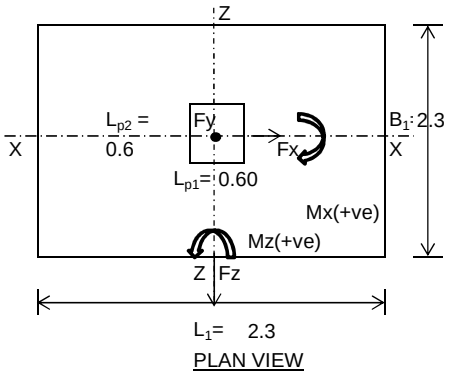
M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.7

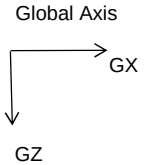
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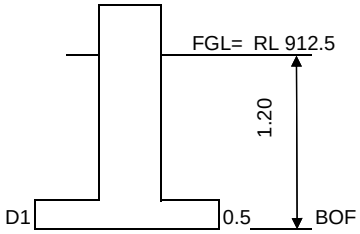
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• Design of Foundation F1



PLAN VIEW





SECTIONAL VIEW

Foundation Level Details :

EL of Finish Ground Level ,FGL =	RL	912.5	m
EL of Highest Water Table ,HWT =	RL	909.5	m
Depth of foundation from FGL. ,H1 =	(-)	1.20	m
Ht of W.T. from FGL level =	(-)	3.00	m
Net Safe Bearing Capacity considered =		450	kN/m^2

Material Properties :

Unit Weight of Soil =		18.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	2.3	m
Footing dimension along Z direction,B_1	=	2.3	m
Depth, D_1	=	0.5	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap = 0.6 x 0.6	=	0.36	m^2
Area of Footing , A = 2.30 x 2.30	=	5.29	m^2
Zz-z = 2.3 x 2.30 x 2.30 / 6	=	2.03	m^3
Zx-x = 2.3 x 2.3 x 2.3 / 6	=	2.03	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	66.1	kN
Wt of soil above ftg =	62.1	kN
Total weight,W=	128	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.6	m
Pedestal dimension along Z direction,Lp2=	0.6	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 1.2 x 18 = 472 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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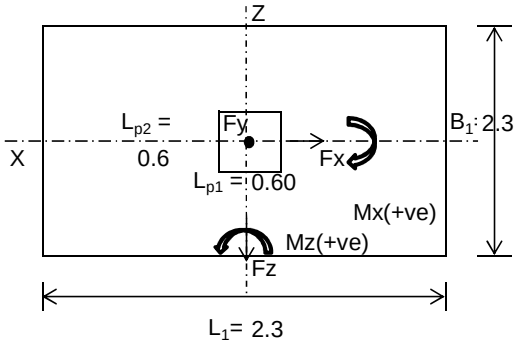
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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F1	Force from Load Combinations						Bearing Pressure Calculation						Check for Overturning:								Check for Sliding					
	Axial Force			Moment									About Z - Axis				About X - Axis									
	Fx	Fy	Fz	Mx	My	Mz	max gross pressure P/A+Mz/Zzz+Mx/Zxx	NSBC increased by	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zzz-Mx/Zxx	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2 + Fz^2}$	P	μ P / Fz	Remarks
L/C	kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	%	kN/m^2		kN/m^2		kN-m	kN-m			kN-m	kN-m			kN	kN	kN	kN		
101	0	-2088	0	0	0	0	394.8	1	471.6	safe	394.8	ok	0.0	2401	-	-	0.0	2401	-	ok	0.0	0.0	0.0	2088	-	ok
102	0	-2088	0	0	0	0	394.8	1	471.6	safe	394.8	ok	0.0	2401	-	-	0.0	2401	-	ok	0.0	0.0	0.0	2088	-	ok
103	0	-2088	0	0	0	0	394.8	1	471.6	safe	394.8	ok	0.0	2401	-	-	0.0	2401	-	ok	0.0	0.0	0.0	2088	-	ok
104	7	-2088	0	0	0	21.7	405.5	1	471.6	safe	384.1	ok	21.7	2401	111	ok	0.0	2401	-	ok	7.0	0.0	7.0	2088	-	ok
105	5.6	-1936	0	0	0	16.7	374.3	1	471.6	safe	357.8	ok	16.7	2227	134	ok	0.0	2227	-	ok	5.6	0.0	5.6	1936	-	ok
106	5.25	-2088	0	0	0	16.3	402.8	1	471.6	safe	386.8	ok	16.3	2401	148	ok	0.0	2401	-	ok	5.3	0.0	5.3	2088	-	ok
107	4.2	-1936	0	0	0	12.5	372.2	1	471.6	safe	359.9	ok	12.5	2227	178	ok	0.0	2227	-	ok	4.2	0.0	4.2	1936	-	ok
108	7	-132	0	0	0	21.7	35.7	1	471.6	safe	14.3	ok	21.7	151	6.99	ok	0.0	151	-	ok	7.0	0.0	7.0	131.7	-	ok
109	5.6	-132	0	0	0	16.7	33.2	1	471.6	safe	16.7	ok	16.7	151	9.1	ok	0.0	151	-	ok	5.6	0.0	5.6	131.7	-	ok

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PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	20	257.7	200	20	244.7	200

Footing Dimension :

Footing dimension along X direction, L_1	=	2.3	m
Footing dimension along Z direction, B_1	=	2.3	m
Depth, D_1	=	0.5	m
Clear cover	=	75	mm

Column Dimension :

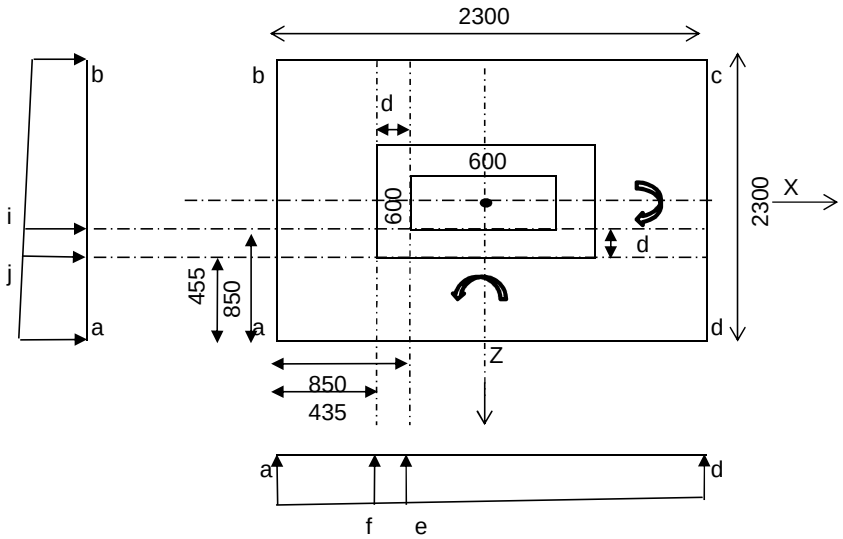
Pedestal dimension along X direction,Lp1	=	0.6	m
Pedestal dimension along Z direction,Lp2	=	0.6	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Critical Load Combination for Design : 201

Maximum Net Base Pressure : 518.7 kN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*518.7]*2.3*0.85*0.85	=	431.0	kN-m
Shear force at Face "f"	= 518.7*0.435*2.3	=	519.0	kN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*518.7]*2.3*0.85*0.85	=	431.0	kN-m
Shear force at Face "j"	= 518.7*0.455*2.3	=	542.8	kN

Design for Bottom Reinforcement :

In X Direction (Along Length 2300 mm)

Cantilever Length		=	850	mm
Dia of Bar		d_b =	20	mm
Effective Depth		d =	415	mm
Bending Moment at "e"		M_u =	431.0	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0061	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y (d-a/2)$	0.9*420*(415-0.0061As/2)As=431*10^6	A_s =	2805.3	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*2300*500	$A_{s,min}$ =	2070	mm^2
Spacing required	2300*22/28*20^2/2805.3	S_{reqd} =	257.7	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2300*22/28*20^2/200	$A_{s,pro}$ =	3614.3	mm^2
$a = A_s f_y / 0.85 f'_c b$	3614.3*420/(0.85*35*2300)	a =	22.19	mm
$c = a/\beta_1$	22.19/0.8	c =	27.74	mm
$\varepsilon_t = \varepsilon_{cu}/c\ (d-c)$	0.003/27.74*(415-27.74)	ε_t =	0.042	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	519.0	kN
Moment at "f"	= [0.5*518.7]*2.3*0.435*0.435	M_u =	112.9	kN-m
ρ_w provided	3614.3/2300/415	$\rho_{w,pro}$ =	0.0038	%
Effective Shear Area	=415*2300/10^6	bd =	0.955	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$		$V_{c,a}$ =	1020.7	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	964.9	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	1637.6	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$		=	3518.9	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 2300 mm)

Cantilever Length		=	850	mm
Dia of Bar		d_b =	20	mm
Effective Depth		d =	395	mm
Bending Moment at "i"		M_u =	431.0	kN-m
C = $0.85f_c'ba$ and T = A_sf_y	Therefore 'a' = $f_y/0.85f_c'b * A_s$	a =	0.0061	$*A_s$
$\varphi M_n \geq M_u = \varphi A_sf_y(d-a/2)$	0.9*420*(395-0.0061As/2)As=430.98*10^6	A_s =	2953.8	mm^2
P min		=	0.18	%
$A_{s,min}$	0.18/100*2300*500	$A_{s,min}$ =	2070	mm^2
Spacing required	2300*22/28*20^2/2953.8	S_{reqd} =	244.72	mm c/c
Spacing Provided		S_{prov} =	200	mm c/c
A_s Provided	2300*22/28*20^2/200	$A_{s,pro}$ =	3614.3	mm^2
$a = A_{st}f_y/0.85f'_cb$	3614.3*420/(0.85*35*2300)	a =	22.18	mm
$c = a/\beta_1$	22.18/0.8	c =	27.73	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/27.73*(395-27.73)	ε_t =	0.040	

One Way Shear Check : > 0.005 tension controlled

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u =	542.8	kN
Moment at "j"	= [0.5*518.7]*2.3*0.455*0.455	M_u =	123.5	kN-m
ρ_w provided	= 3614.3/2300/395	$\rho_{w,pro}$ =	0.0040	%
Effective Shear Area	=395*2300/10^6	bd =	0.909	m^2
V_c = lesser of				
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_wV_ud/M_u)bd$		$V_{c,a}$ =	966.6	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$ =	921.4	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$ =	1558.7	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$		=	3351.6	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok			ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$ =	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure		=	518.70	kN/m^2
Shear force,V_u	= 518.7*[2.3*2.3-(0.6+0.415)*(0.6+0.395)]	=	2220.08	kN
Shear peremeter,b_0	= 2*[(0.6+0.415)+(0.6+0.395)]	=	4.02	m

$V_c = 0.17(1+2/\beta)\sqrt{f_c'}b_0d$ = 0.51 $\sqrt{f_c'}b_0d$

$V_c = 0.083(\alpha_sd/b_0 + 2)\sqrt{f_c'}b_0d$ = 0.42 $\sqrt{f_c'}b_0d$ — For Edge column, α_s = 30

$V_c = 0.33\sqrt{f_c'}b_0d$ = 0.33 $\sqrt{f_c'}b_0d$

φV_c	= 0.75*0.33*SQRT(35)*4020*415/1000	=	2442.8	kN
Remarks : if $\varphi V_c > V_u$ ok	= safe			

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- Outer Column
- Load Calculation :

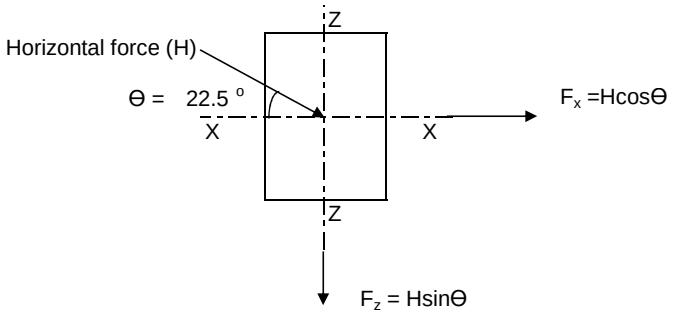
Input load data for Neurtalization thickener

Load Table :

	Tank dead load	Max fluid load		Normal operating fluid load		Seismic load			Wind load			Bridge/drive dead load	Bridge/drive live load
Outer Column	kN	kN		kN		kN	kN	Kn-m	kN	kN	kN-m	kN	kN
	V	H	V	H	V	H	V	Moment	H	V	Moment	V	V
	13	34	214	31	197	6	8	0	5	11	0	25	68

H : Horizontal load V : Vertical load

Horizontal reactions are in the plane of bracing.



Depth of foundation from U/S of base plate = 2.1 m

Load at bottom of foundation level

Primary load cases		F_x	F_y	F_z	M_x	M_z
		kN	kN	kN	kN-m	kN-m
Tank dead load	D	0	13	0	0	0
Selfweight of column = 0.4*0.4*(1.6-0.3)*25 =		0	5.2	0	0	0
Selfweight of footing =		0	27.075	0	0	0
Weight of soil above footing =		0	56	0	0	0
Bridge/drive dead load		0	25	0	0	0
Max fluid load	F_{max}	31.41	214	13.02	27.342	65.961
Normal operating fluid load	$F_{operati}$	28.64	197	11.87	24.927	60.144
Seismic load	E	5.55	8	2.3	4.83	11.655
Wind load	W	4.62	11	1.92	4.032	9.702
Bridge/drive live load	L	0	68	0	0	0

Combinations including seismic load to use normal operating fluid load.

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Load combinations for foundation servicibility check :

Sl no	Basic combinations as per cl 2.4.1 SBC 301	F_x	F_y	F_z	M_x	M_y	M_z
		kN	kN	kN	kN-m	kN-m	kN-m
101	$D+F_{max}$	31.41	340	13.02	27.34	0	66
102	$D+F_{max}+L$	31.41	408	13.02	27.34	0	66
103	$D+F_{max}+0.75L$	31.41	391	13.02	27.34	0	66
104	$D+F_{max}+W$ (+ve)	36.03	351	14.94	31.37	0	75.7
105	$D+F_{operating}+0.7E$ (+ve)	32.53	329	13.48	28.31	0	68.3
106	D+F+0.75L+0.75W (+ve)	34.88	399	14.46	30.37	0	73.2
107	$D+F_{operation}+0.75L+0.75*(0.7E)$ (+ve)	31.55	378	13.08	27.46	0	66.3
108	0.6D+W (+ve)	4.62	109	1.92	4.032	0	9.7
109	0.6D+0.7E (+ve)	3.885	104	1.61	3.381	0	8.16

Weight of soil above footing includes in all load combination except (0.6D+W or 0.6D+E).

Load combinations for foundation strength check :

Sl no	Basic combinations as per cl 2.3.2 SBC 301	F_x	F_y	F_z	M_x	M_y	M_z	q_{net1}	q_{net2}	q_{net3}	q_{net4}	q_{avg}
		kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	kN/m^2	kN/m^2	kN/m^2	kN/m^2
201	$1.4(D+F_{max})$	44	360	18.2	38.28	0	92.35	214	52.4	147	-14.5	181
202	$1.4(D+F_{max})+1.7L$	44	476	18.2	38.28	0	79.15	234.5	96	168	29	201
203	1.2D+f1L	0	120	0	0	0	0	33.2	33.2	33.2	33.2	33.2
204	1.2D+0.8W (+ve)	3.7	61	1.54	3.226	0	7.762	26.41	12.8	20.8	7.19	23.6
205	1.2D+1.6W (+ve) +f1L	7.39	137	3.07	6.451	0	15.52	57.29	30.1	46	18.8	51.7
206	1.2D+1.0E (+ve) +f1L	5.55	128	2.3	4.83	0	11.66	49.83	29.4	41.4	21	45.6
207	0.9D+1.6W (+ve)	7.39	56.48	3.07	6.451	0	15.52	34.87	7.71	23.6	-3.58	29.2
208	0.9D+1.0E (+ve)	5.55	46.88	2.3	4.83	0	11.66	27.41	7.02	19	-1.43	23.2

f_1 = 1 cl 2.4.1 SBC 301

Design Parameter for net base pressure calculation:

Area of Footing , A = 3.61 m^2

Zz-z = 1.14 m^3

Zx-x = 1.14 m^3

Net Pressure, $q_{net} = P/A \pm M_z/Z_{zz} \pm M_x/Z_{xx}$

Consider average net pressure for design.

q_{avg} for design = 201 kN/m^2

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• Design of Column C2 :

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1.2 % i.e. ρ_g = 0.012

β = 0.65

k = 2

Column Dimensions :

Depth parallel to X-X axis, h = 400 mm

Depth parallel to Z-Z axis, b = 400 mm

Clear Cover = 75 mm

d' = 95 mm

Y_h = (h-d')/h = 0.53

Y_b = (b-d')/b = 0.53

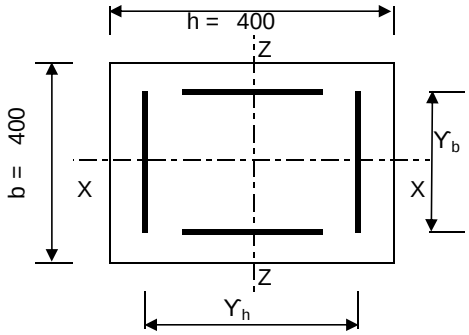
Unsupported length of column parallel to X-X axis, l_{ux} = 1.3 m kl_{ux}/b = 6.5 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.3 m kl_{uz}/h = 6.5

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 4452 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	kN	kN-m		kN	kN-m				kN-m			kN-m				
201	360.1	38.3	92.3	554.0	58.9	142.1	0.099	0.075	168	0.35	0.075	168	0.85	1.61	0.95	safe
202	475.7	38.3	79.2	731.8	58.9	121.8	0.131	0.075	168	0.35	0.075	168	0.72		0.78	safe
203	119.8	0.0	0.0	184.4	0.0	0.0	0.033	0.070	156.8	0.00	0.070	156.8	0.00		0	safe
204	60.6	3.2	7.8	93.3	5.0	11.9	0.017	0.070	156.8	0.03	0.070	156.8	0.08		0.02	safe
205	137.4	6.5	15.5	211.4	9.9	23.9	0.038	0.070	156.8	0.06	0.070	156.8	0.15		0.06	safe
206	127.8	4.8	11.7	196.7	7.4	17.9	0.035	0.070	156.8	0.05	0.070	156.8	0.11		0.04	safe
207	56.5	6.5	15.5	86.9	9.9	23.9	0.016	0.070	156.8	0.06	0.070	156.8	0.15		0.06	safe
208	46.9	4.8	11.7	72.1	7.4	17.9	0.013	0.070	156.8	0.05	0.070	156.8	0.11		0.04	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
1920	4	20	+	4	16	2062	1.29	10	256	250



P_n : Nominal axial load P_u : Factored design axial load

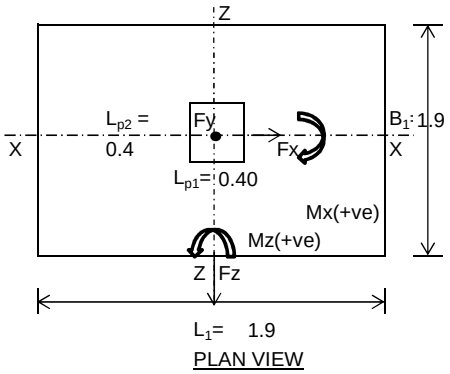
M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.6

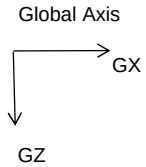
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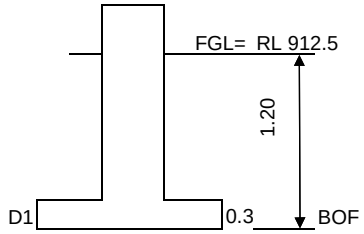
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• **Design of Foundation F2**



PLAN VIEW





SECTIONAL VIEW

Foundation Level Details :

EL of Finish Ground Level ,FGL =	RL	912.5	m
EL of Highest Water Table ,HWT =	RL	909.5	m
Depth of foundation from FGL. ,H1 =	(-)	1.20	m
Ht of W.T. from FGL level =	(-)	3.00	m
Net Safe Bearing Capacity considered =		450	kN/m^2

Material Properties :

Unit Weight of Soil =		18.0	kN/m^3
Concrete Density =		25.0	kN/m^3
Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa
Concrete to soil coefficient of friction, μ =		0.4	
FOS against Sliding	=	1.5	
FOS against Overturning	=	2	DL
FOS against Overturning	=	1.5	DL+LL

Footing Dimension :

Footing dimension along X direction,L_1	=	1.9	m
Footing dimension along Z direction,B_1	=	1.9	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Design Parameter :

Area of Pedestal , Ap = 0.4 x 0.4	=	0.16	m^2
Area of Footing , A = 1.90 x 1.90	=	3.61	m^2
Zz-z = 1.9 x 1.90 x 1.90 / 6	=	1.14	m^3
Zx-x = 1.9 x 1.9 x 1.9 / 6	=	1.14	m^3

Foundation Weight

	Dry Condn	
Self weight of Footing =	27.1	kN
Wt of soil above ftg =	55.9	kN
Total weight,W=	83	kN

Column Dimension :

Pedestal dimension along X direction,Lp1=	0.4	m
Pedestal dimension along Z direction,Lp2=	0.4	m

Bearing Capacity :

GBC @ Bottom of footing = NSBC + 1.2 x 18 = 472 kN/m^2

GBC : Gross bearing capacity
NSBC : Net safe bearing capacity

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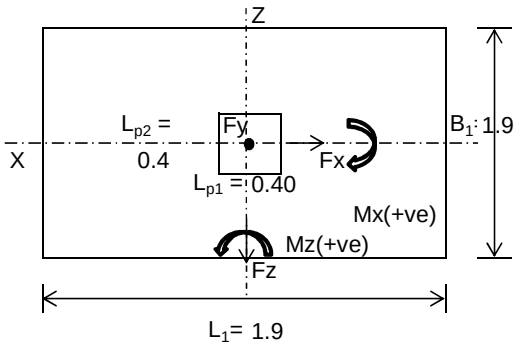
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Check for allowable base pressure, overturning & sliding of foundation

Foundation Mkd. F2	Force from Load Combinations						Bearing Pressure Calculation						Check for Overturning:								Check for Sliding					
	Axial Force			Moment			max gross pressure P/A+Mz/Zzz+Mx/Zxx	NSBC increased by	Allowable Bearing Capacity	Remarks	Min Gross Pressure P/A-Mz/Zzz-Mx/Zxx	Remarks	About Z - Axis				About X - Axis										
	Fx	Fy	Fz	Mx	My	Mz							Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Overturning Moment, M_o	Resisting Moment, M_r	Mr / Mo	Remarks	Fx	Fz	$\sqrt{Fx^2 + Fz^2}$	P	μ P / Fz	Remarks
L/C	kN	kN	kN	kN-m	kN-m	kN-m	kN/m^2	%	kN/m^2		kN/m^2		kN-m	kN-m			kN-m	kN-m			kN	kN	kN	kN		
101	31.41	-340	13	27.3	0	66	175.9	1	471.6	safe	12.7	ok	66.0	323	4.9	ok	27.3	323	11.8	ok	31.4	13.0	34.0	340.2	10.5	ok
102	31.41	-408	13	27.3	0	66	194.7	1	471.6	safe	31.5	ok	66.0	388	5.88	ok	27.3	388	14.2	ok	31.4	13.0	34.0	408.2	12.5	ok
103	31.41	-391	13	27.3	0	66	190.0	1	471.6	safe	26.8	ok	66.0	372	5.64	ok	27.3	372	13.6	ok	31.4	13.0	34.0	391.2	12	ok
104	36.03	-351	14.9	31.4	0	75.7	191.0	1	471.6	safe	3.7	ok	75.7	334	4.41	ok	31.4	334	10.6	ok	36.0	14.9	39.0	351.2	9.41	ok
105	32.53	-329	13.5	28.3	0	68.3	175.6	1	471.6	safe	6.6	ok	68.3	312	4.58	ok	28.3	312	11	ok	32.5	13.5	35.2	328.8	9.76	ok
106	34.88	-399	14.5	30.4	0	73.2	201.3	1	471.6	safe	20.1	ok	73.2	379	5.19	ok	30.4	379	12.5	ok	34.9	14.5	37.8	399.4	11.1	ok
107	31.55	-378	13.1	27.5	0	66.3	186.8	1	471.6	safe	22.9	ok	66.3	359	5.43	ok	27.5	359	13.1	ok	31.6	13.1	34.2	378.4	11.6	ok
108	4.62	-109	1.92	4.03	0	9.7	42.3	1	471.6	safe	18.2	ok	9.7	104	10.7	ok	4.0	104	25.7	ok	4.6	1.9	5.0	109.1	22.7	ok
109	3.885	-104	1.61	3.38	0	8.16	38.9	1	471.6	safe	18.7	ok	8.2	98	12.1	ok	3.4	98.5	29.1	ok	3.9	1.6	4.2	103.7	25.8	ok

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PLAN VIEW

Reinforcement Summary :

Bottom Reinforcement	In X dirn			In Z dirn		
	Bar Dia	Spacing required	Spacing provided	Bar Dia	Spacing required	Spacing
	16	284.9	250	16	262.9	250

Footing Dimension :

Footing dimension along X direction, L_1	=	1.9	m
Footing dimension along Z direction, B_1	=	1.9	m
Depth, D_1	=	0.3	m
Clear cover	=	75	mm

Column Dimension :

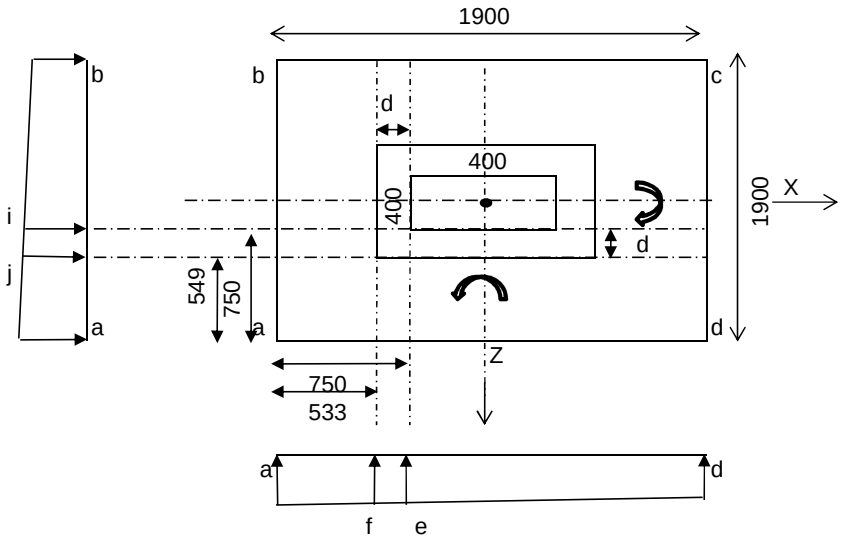
Pedestal dimension along X direction,Lp1	=	0.4	m
Pedestal dimension along Z direction,Lp2	=	0.4	m

Design Constants :

φ for Moment =	=	0.9
φ for shear =	=	0.75
β_1	=	0.8
ε_t limit (for tension controlled section) =		0.005
Concrete compressive strain, ε_{cu}	=	0.003

Material Properties :

Concrete Grade =	f_c'	35	Mpa
Reinforcement Grade =	f_y	420	Mpa



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Critical Load Combination for Design : 201

Maximum Net Base Pressure : 201.1 KN/m^2

Moment & Shear Calculation Along X dirn :

Design Moment at Face "e"	= [0.5*201.1]*1.9*0.75*0.75	=	107.5	kN-m
Shear force at Face "f"	= 201.1*0.533*1.9	=	203.7	kN

Moment & Shear Calculation Along Z dirn :

Design Moment at Face "i"	= [0.5*201.1]*1.9*0.75*0.75	=	107.5	kN-m
Shear force at Face "j"	= 201.1*0.549*1.9	=	209.8	kN

Design for Bottom Reinforcement :

In X Direction (Along Length 1900 mm)

Cantilever Length		=	750	mm
Dia of Bar		d_b =	16	mm
Effective Depth		d =	217	mm
Bending Moment at "e"		M_u =	107.5	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a =	0.0074	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y (d-a/2)$	0.9*420*(217-0.0074As/2)As=107.5*10^6	A_s =	1341.2	mm^2
P min	[SBC 304 Table 7.6.1.1]	=	0.18	%
$A_{s,min}$ =	0.18/100*1900*300	$A_{s,min}$ =	1026	mm^2
Spacing required	1900*22/28*16^2/1341.2	S_{reqd} =	284.9	mm c/c
Spacing Provided		S_{prov} =	250	mm c/c
A_s Provided	1900*22/28*16^2/250	$A_{s,pro}$ =	1528.7	mm^2
a = $A_s f_y$/0.85f'$_c$b	1528.7*420/(0.85*35*1900)	a =	11.36	mm
c = a/β_1	11.36/0.8	c =	14.20	mm
$\varepsilon_t = \varepsilon_{cu}$/c (d-c)	0.003/14.2*(217-14.2)	ε_t =	0.043	

> 0.005 tension controlled

One Way Shear Check :

Critical section at a distance d from the face of column.

Shear Force at "f"		V_u =	203.7	kN
Moment at "f"	= [0.5*201.1]*1.9*0.533*0.533	M_u =	54.3	kN-m
ρ_w provided	1528.7/1900/217	$\rho_{w,pro}$ =	0.0037	%
Effective Shear Area	=217*1900/10^6	bd =	0.412	m^2

V_c = lesser of

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$	$V_{c,a}$ =	411.4	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$	$V_{c,b}$ =	416.3	kN
c) $0.29\lambda\sqrt{f_c'}bd$	$V_{c,c}$ =	707.4	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd)$	=	1516.0	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok		ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required	$A_{v,min}$ =	Not required	

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In Z Direction (Along Length 1900 mm)

Cantilever Length			=	750	mm
Dia of Bar		d_b	=	16	mm
Effective Depth		d	=	201	mm
Bending Moment at "i"		M_u	=	107.5	kN-m
C = 0.85f_c'ba and T = $A_s f_y$	Therefore 'a' = f_y/0.85f_c'b * A_s	a	=	0.0074	*A_s
$\varphi M_n \geq M_u = \varphi A_s f_y(d-a/2)$	0.9*420*(201-0.0074As/2)As=107.47*10^6	A_s	=	1453.4	mm^2
P min			=	0.18	%
$A_{s,min}$	0.18/100*1900*300	$A_{s,min}$	=	1026	mm^2
Spacing required	1900*22/28*16^2/1453.4	S_{reqd}	=	262.95	mm c/c
Spacing Provided		S_{prov}	=	250	mm c/c
A_s Provided	1900*22/28*16^2/250	$A_{s,pro}$	=	1528.7	mm^2
$a = A_{st} f_y/0.85f'_c b$	1528.7*420/(0.85*35*1900)	a	=	11.36	mm
$c = a/\beta_1$	11.36/0.8	c	=	14.20	mm
$\varepsilon_t = \varepsilon_{cu}/c$ (d-c)	0.003/14.2*(201-14.2)	ε_t	=	0.039	

One Way Shear Check : > 0.005 tension controlled

Critical section at a distance d from the face of column.

Shear Force at "j"		V_u	=	209.8	kN
Moment at "j"	= [0.5*201.1]*1.9*0.549*0.549	M_u	=	57.6	kN-m
ρ_w provided	= 1528.7/1900/201	$\rho_{w,pro}$	=	0.0040	%
Effective Shear Area	=201*1900/10^6	bd	=	0.382	m^2

V_c = lesser of

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$		$V_{c,a}$	=	380.5	kN
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$		$V_{c,b}$	=	387.5	kN
c) $0.29\lambda\sqrt{f_c'}bd$		$V_{c,c}$	=	655.2	kN
$\varphi(V_c + 0.66\sqrt{f_c'}bd$			=	1403.8	kN
Remarks : if $V_u \leq \varphi(V_c + 0.66\sqrt{f_c'}bd)$ ok				ok	
Remarks : if $V_u > \varphi V_c$, minimum area of shear reinforcement required		$A_{v,min}$	=	Not required	

Two Way Shear Check :

Critical section at a distance d/2 from the face of column.

β = short side of column / long side of column = 1.00

Average soil pressure			=	201.10	kN/m^2
Shear force,V_u	= 201.1*[1.9*1.9-(0.4+0.217)*(0.4+0.201)]		=	651.40	kN
Shear peremeter,b_0	= 2*[(0.4+0.217)+(0.4+0.201)]		=	2.44	m
$V_c = 0.17(1+2/\beta)\sqrt{f_c'}b_0d$	= 0.51 $\sqrt{f_c'}b_0d$				
$V_c = 0.083(\alpha_s d/b_0 + 2)\sqrt{f_c'}b_0d$	= 0.39 $\sqrt{f_c'}b_0d$	For Edge column	α_s	30	
$V_c = 0.33\sqrt{f_c'}b_0d$	= 0.33 $\sqrt{f_c'}b_0d$				
φV_c	= 0.75*0.33*SQRT(35)*2436*217/1000		=	774.0	kN
Remarks : if $\varphi V_c > V_u$ ok	= safe				

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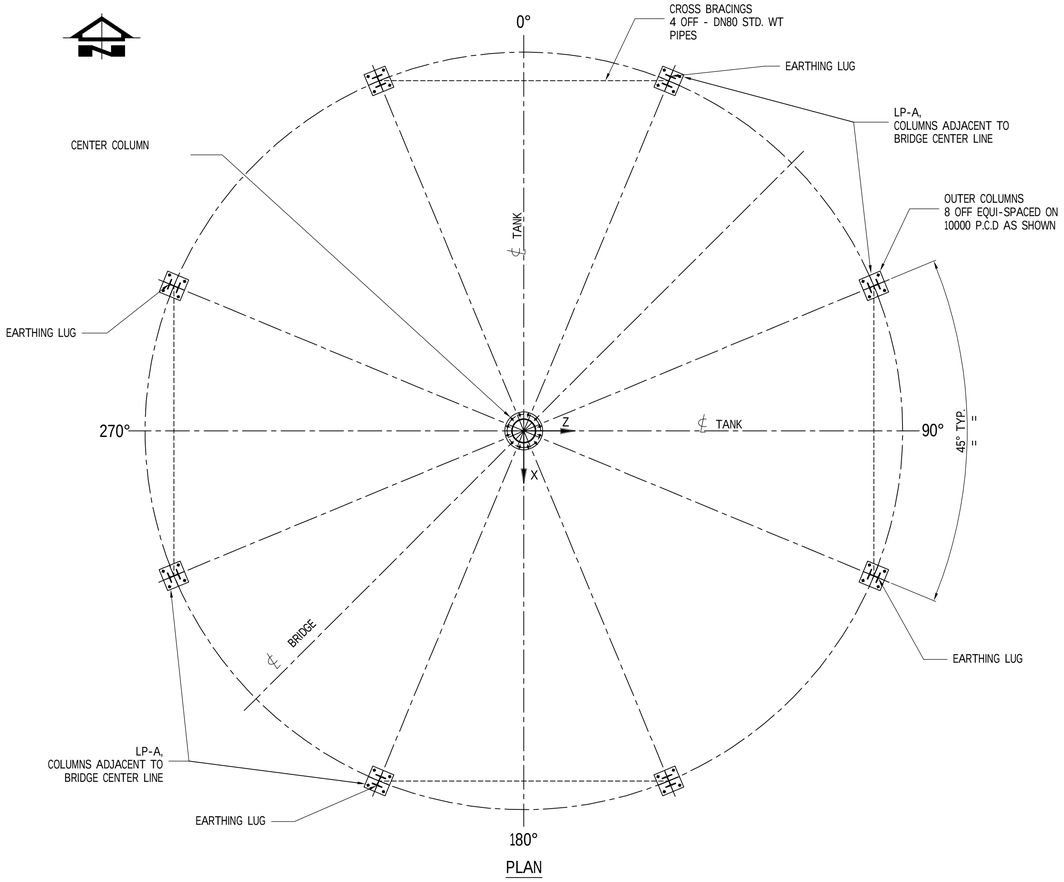
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ATTACHMENT 1 OTE INPUT DRAWINGS(FOOTING AND LOADING PLAN)

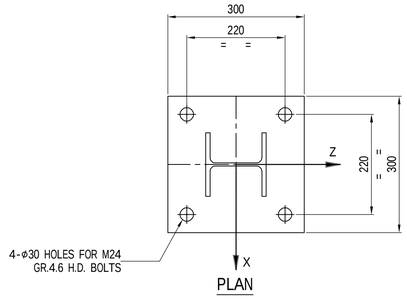
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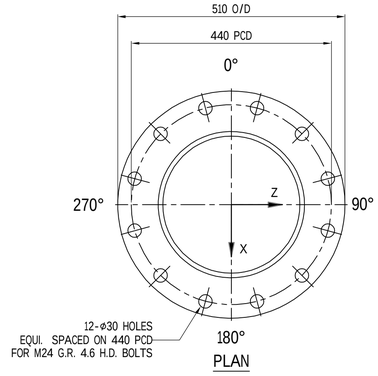
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REV: A01



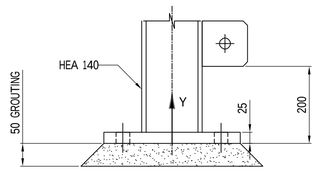
PLAN



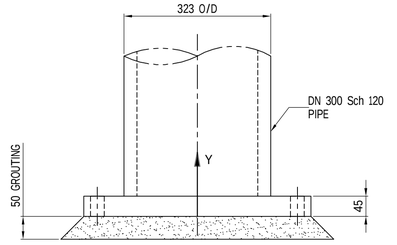
PLAN



PLAN



ELEVATION
OUTER COLUMN
SCALE 1:6



ELEVATION
CENTRE COLUMN
SCALE 1:6

LOAD TABLE

		Tank Dead Load	Max Fluid Load SG 1.204		Normal Operating Fluid Load SG 1.108		Seismic Load (+/-)			Wind Load (+/-)			Bridge/Drive Dead Load (LP-A) *NOTE 9	Bridge/Drive Live Load (LP-A) *NOTE 9
	Column PCD (mm)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Horizontal (X,Z) (kN)	Vertical (Y) (kN)	Moment (X,Z) (kNm)	Vertical (Y) (kN)	Vertical (Y) (kN)
Outer Columns	10 000	13	34	214	31	197	6	8	0	5	11	0	25	68
Central Column	0	40	0	1 910	0	1 758	8	0	7	7	0	7	0	0
Total Applied Load	-	144	0	3 622	0	3 333	32	0	109	25	0	99	73	170

NOTES:
1. ALL LOADS LISTED ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301
2. SEISMIC LOADS ARE CALCULATED CONSIDERING DYNAMIC SLOSHING EFFECTS OF NORMAL OPERATING SG LIQUIDS
3. COMBINATIONS INCLUDING SEISMIC LOAD TO USE NORMAL OPERATING FLUID LOAD
4. ALL HORIZONTAL REACTIONS ARE IN THE PLANE OF BRACING
5. TOTAL LOADS AND MOMENTS ARE CALCULATED AT THE UNDERSIDE OF BASEPLATE
6. COLUMN LOADS ARE GIVEN PER COLUMN
7. DIFFERENTIAL SETTLEMENT BETWEEN ANY ADJACENT COLUMN SHALL BE LIMITED TO L / 600
8. TILTING OF THE TANK DUE TO OUTER COLUMN DIFFERENTIAL SETTLEMENT SHALL BE LIMITED TO 8 mm
9. BRIDGE/DRIVE LOADS REACTED AT LOAD POINTS A (LP-A) ONLY. LP-A IS ADJACENT TO BRIDGE CENTER LINE.
10. LENGTH OF ANCHOR BOLTS TO BE DEFINED AS A FUNCTION OF CIVIL ENGINEERING

SEISMIC DATA TO SBC301

Ss, S1:	0.10, 0.02	
Importance Factor:	1	
Site Class:	B	

WIND DATA TO SBC301

V:	43	m/s
Exposure Class:	C	

FOUNDATION TOLERANCES

HEIGHT OF COLUMN FOUNDATIONS	± 10 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT IN A GROUP	± 3 mm
ANCHOR BOLTS - DISTANCE BETWEEN ANY BOLT GROUP	± 6 mm
ANCHOR BOLTS - MAX TOLERANCE ACCUMULATION BETWEEN BOLT GROUPS	± 6 mm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-ME-DED-1001	GENERAL NOTES								
MD-216-3000-EG-ME-SPC-3001	TECHNICAL SPECIFICATION								
MD-216-3520-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3520-EG-ME-GAD-1001	EQUIPMENT GENERAL ARRANGEMENT DRAWING	A01	14.11.2019	ISSUED FOR APPROVAL	TK	AC	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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OUTOTEC DRAWING NUMBER: 900707 _GAU01 _RN13 _MLD15 _35201 — DOCUMENT ID: OU601250817 — REVISION: 1

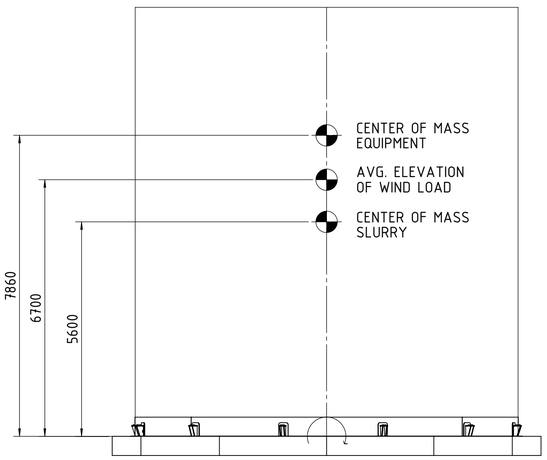
DRAWN	timkar	13.11.2019
CHECKED	alecha	13.11.2019
ENGINEER	soucha	14.11.2019
APPROVED	ricleh	14.11.2019
EMPLOYER		
SCALE	1:40	A1

MANSOURAH-MASSARAH GOLD PROJECT
FOOTING AND LOAD PLAN
NEUTRALIZATION AND THICKENING
3520-THK-001 T80-NEUTRALIZATION THK, 12m

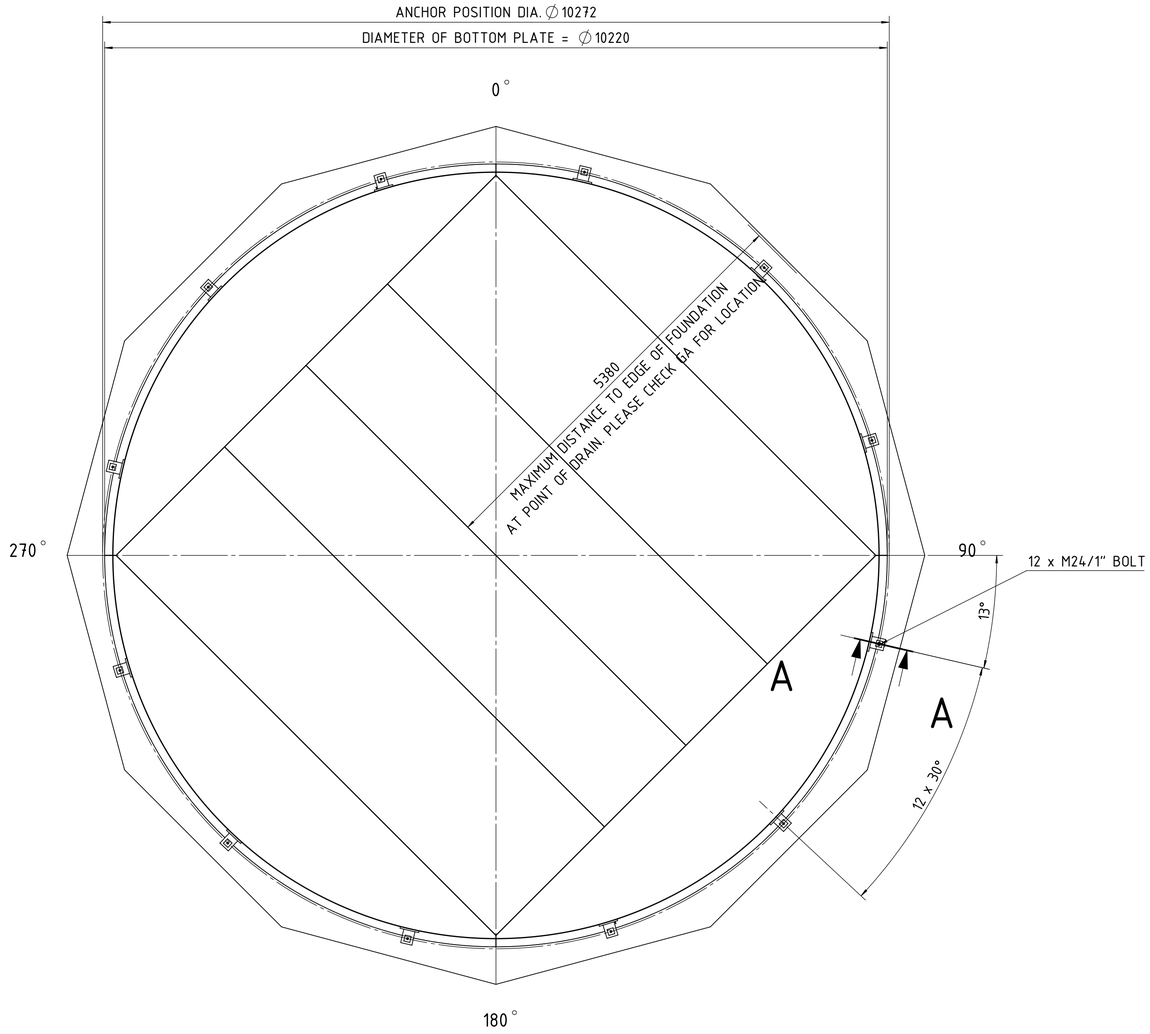
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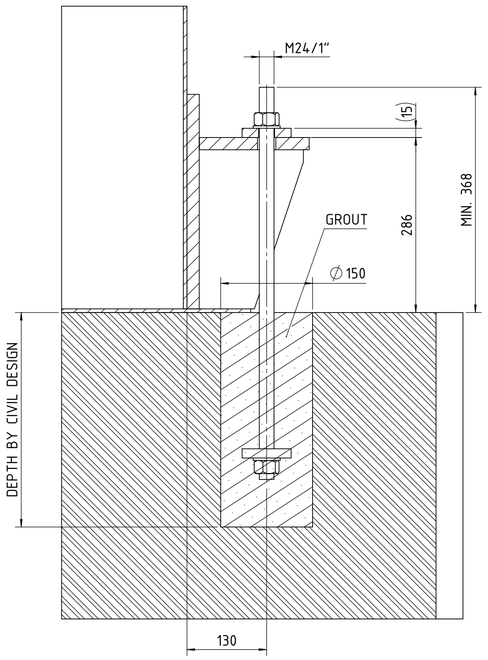
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REV: A01

LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 80





SECTION A-A
SCALE 1 : 5

NEUTRALIZATION THICKENER UNDERFLOW TANK	
DEAD WEIGHT	44150 kg
OPERATIONAL WEIGHT OF SLURRY	952700 kg
MAXIMUM WEIGHT OF SLURRY	1066900 kg
EXTERNAL PLATFORM LOAD	4390 kg
NORMAL OPERATING SPEED	28 rpm
WIND LOAD	106 kN
SEISMIC LOAD	436 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	2481 kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3520-EG-ME-GAD-1011	GENERAL ARRANGEMENT DRAWING								
MD-216-3520-EG-ME-DAT-1003	EQUIPMENT DATASHEET								
MD-216-3520-EG-PR-PID-1022	P&I DIAGRAM								
MD-216-3520-EG-PR-PFD-1003	PROCESS FLOW DIAGRAM	A01	20.12.2019	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-D-01204-007 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707 GAU01 RN13 MLD15 35208 — DOCUMENT ID: OU601266758 — REVISION: 2

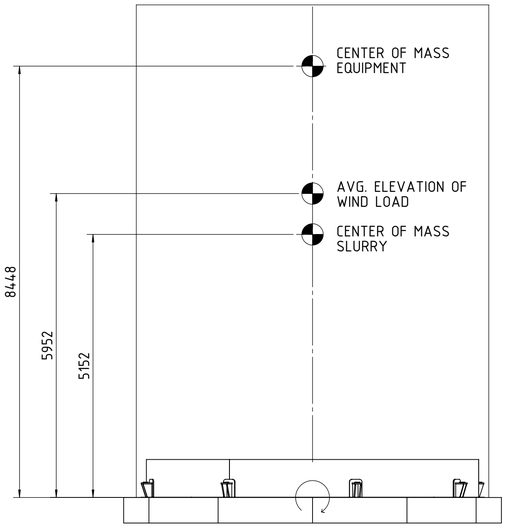
DRAWN	joujaa	20.12.2019
CHECKED	tuoman	20.12.2019
ENGINEER	juhris	20.12.2019
APPROVED	GeoRie	20.12.2019
EMPLOYER		
SCALE	1:40	A1

MANSOURAH-MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
NEUTRALIZATION
3520-REA-007 NEUTRALIZATION THICKENER UNDERFLOW TANK

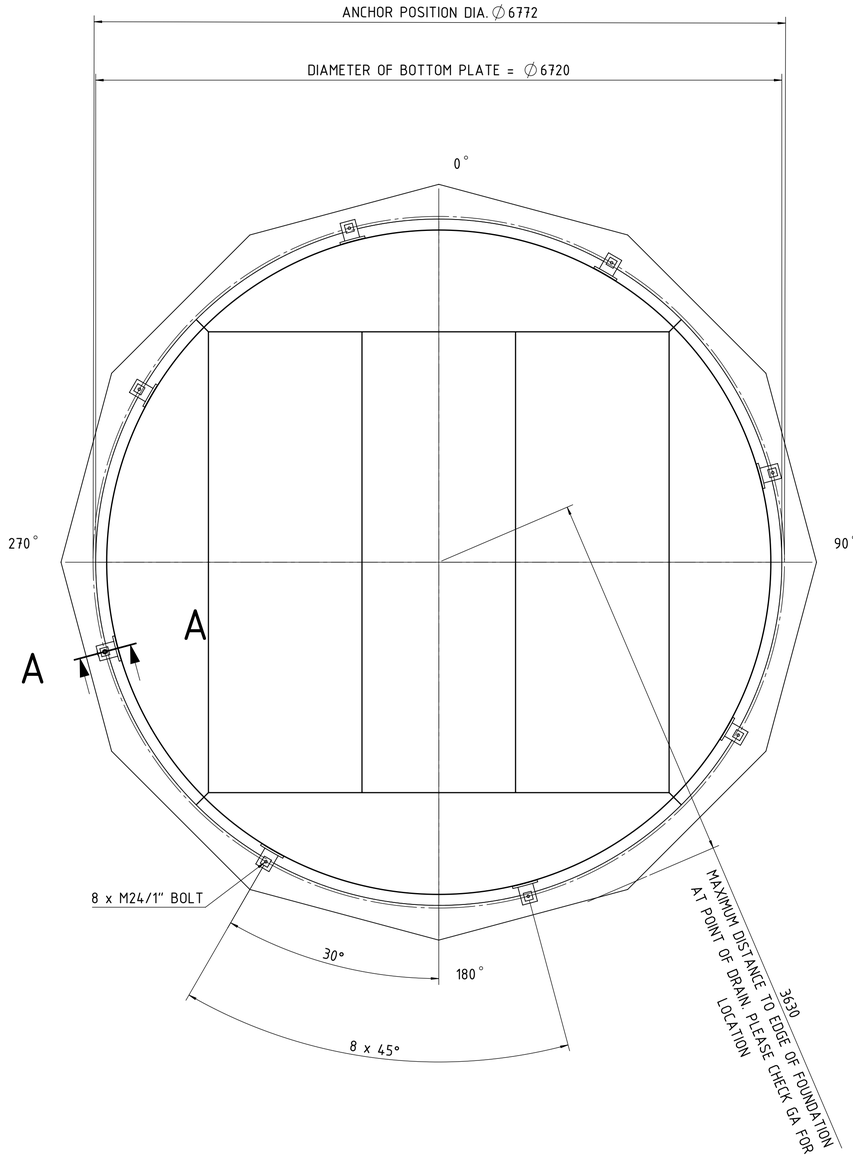
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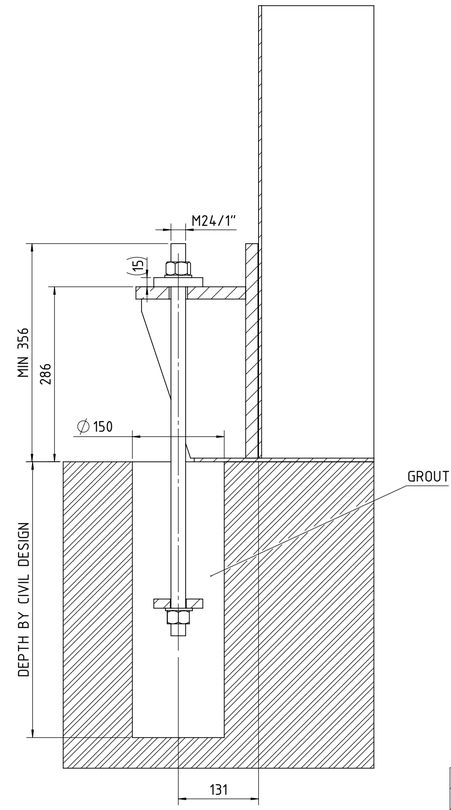
DRAWING NUMBER	REV
MD-216-3520-EG-ME-PLN-1001	A01

LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 60





SECTION A-A
SCALE 1 : 5

NEUTRALIZATION REACTOR	
DEAD WEIGHT	29900 kg
OPERATIONAL WEIGHT OF SLURRY	355600 kg
MAXIMUM WEIGHT OF SLURRY	384700 kg
EXTERNAL PLATFORM LOAD	3100 kg
NORMAL OPERATING SPEED	45,7 rpm
WIND LOAD	68 kN
SEISMIC LOAD	162kN
SEISMIC TORQUE AT FOUNDATION LEVEL	877kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

PLANT NORTH

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3520-EG-ME-GAD-1004, -06, -08	GENERAL ARRANGEMENT								
MD-216-3520-EG-ME-DAT-1004...1005	EQUIPMENT DATASHEET								
MD-216-3520-EG-PR-PID-1001, 03, 05	P&I DIAGRAM								
MD-216-3520-EG-PR-PFD-1001...1002	PROCESS FLOW DIAGRAM	A01	05.12.2019	ISSUED FOR APPROVAL	MH	TM	JR	GR	

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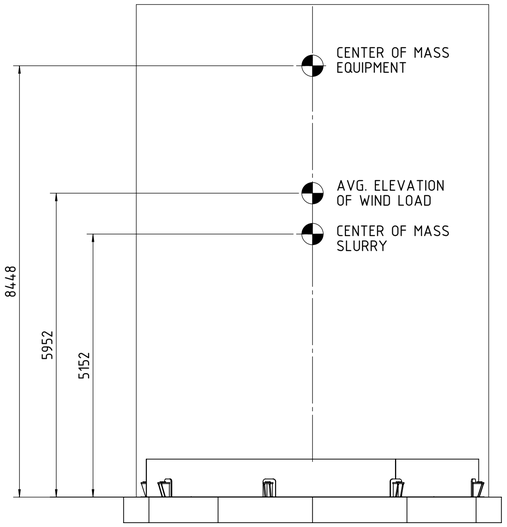
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LARSEN & TOUBRO DRAWING NUMBER	O19045-450-M-D-01204-001		REVISION 2
OUTOTEC DRAWING NUMBER	900707 GAU01 RN13 MLD15 35202	DOCUMENT ID OU601266873	REVISION 2

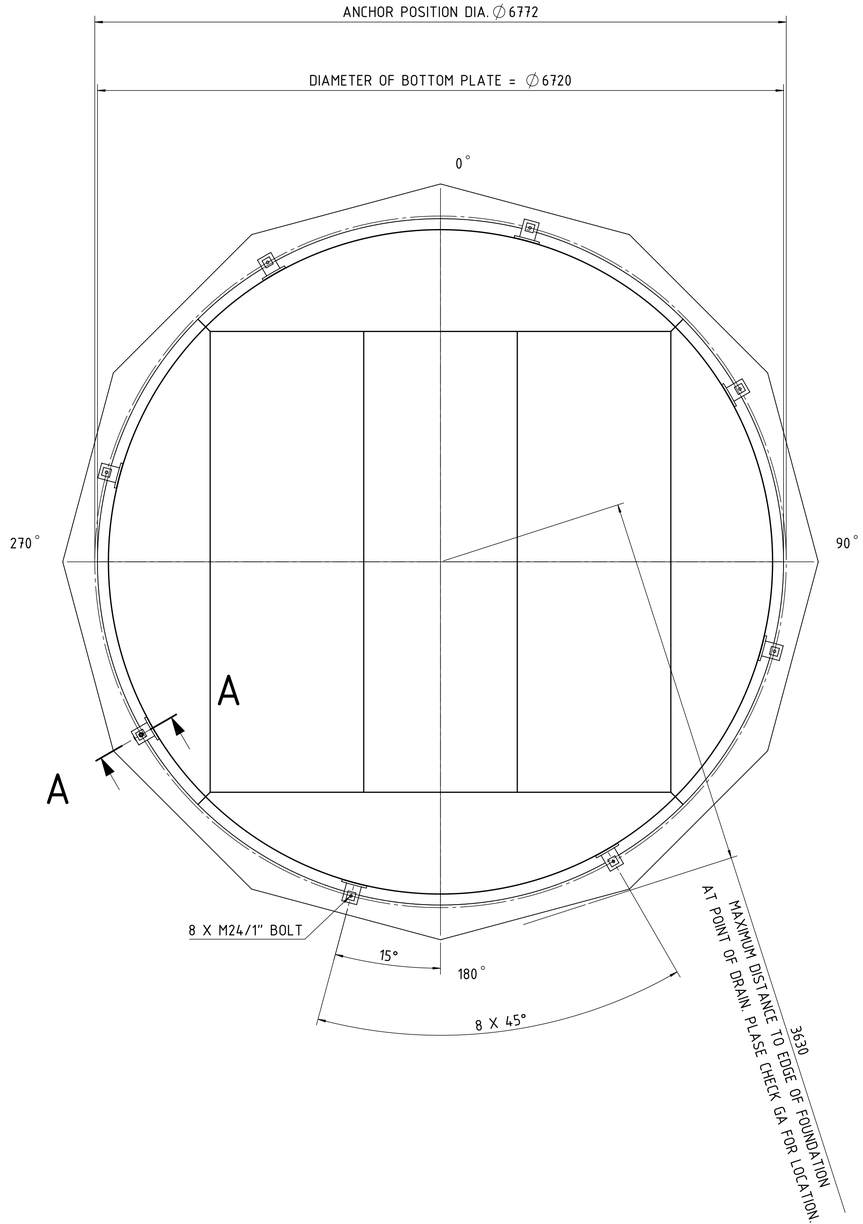
DRAWN	mikhov	05.12.2019	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	tuoman	05.12.2019	FOUNDATION AND LOAD PLAN
ENGINEER	juhris	17.12.2019	NEUTRALIZATION
APPROVED	GeoRie	17.12.2019	3520-REA-001,-003,-005 NEUTRALIZATION REACTOR
EMPLOYER			
SCALE	1:30	A1	DRAWING NUMBER MD-216-3520-EG-ME-PLN-1001 REV A01

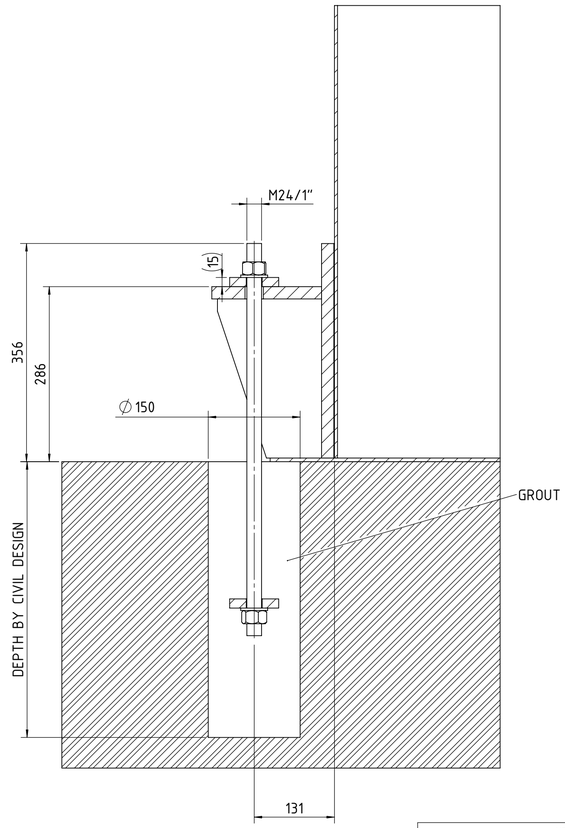
DRAWING NUMBER: MD-216-3520-EG-ME-PLN-1002
REV: A01

LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 60





SECTION A-A
SCALE 1 : 5

NEUTRALIZATION REACTOR	
DEAD WEIGHT	29900 kg
OPERATIONAL WEIGHT OF SLURRY	355600 kg
MAXIMUM WEIGHT OF SLURRY	384700 kg
EXTERNAL PLATFORM LOAD	3100 kg
NORMAL OPERATING SPEED	45,7 rpm
WIND LOAD	68 kN
SEISMIC LOAD	162 kN

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

PLANT NORTH

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3520-EG-ME-GAD-1005, -07, -09	GENERAL ARRANGEMENT								
MD-216-3520-EG-ME-DAT-1004...1005	EQUIPMENT DATASHEET								
MD-216-3520-EG-PR-PID-1002, -04, -06	P&I DIAGRAM								
MD-216-3520-EG-PR-PFD-1001...1002	PROCESS FLOW DIAGRAM	A01	05.12.2019	ISSUED FOR APPROVAL	MH	TM	JR	GR	

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OUTOTEC DRAWING NUMBER: 900707 GAU01 RN13 MLD15 35203 — DOCUMENT ID: OU601267010 — REVISION: 2

DRAWN	mikhov	05.12.2019
CHECKED	tuoman	17.12.2019
ENGINEER	juhris	17.12.2019
APPROVED	GeoRie	17.12.2019
EMPLOYER		
SCALE	1:30	A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
NEUTRALIZATION
3520-REA-002,-004,-006 NEUTRALIZATION REACTOR

DRAWING NUMBER: MD-216-3520-EG-ME-PLN-1002 — REV: A01

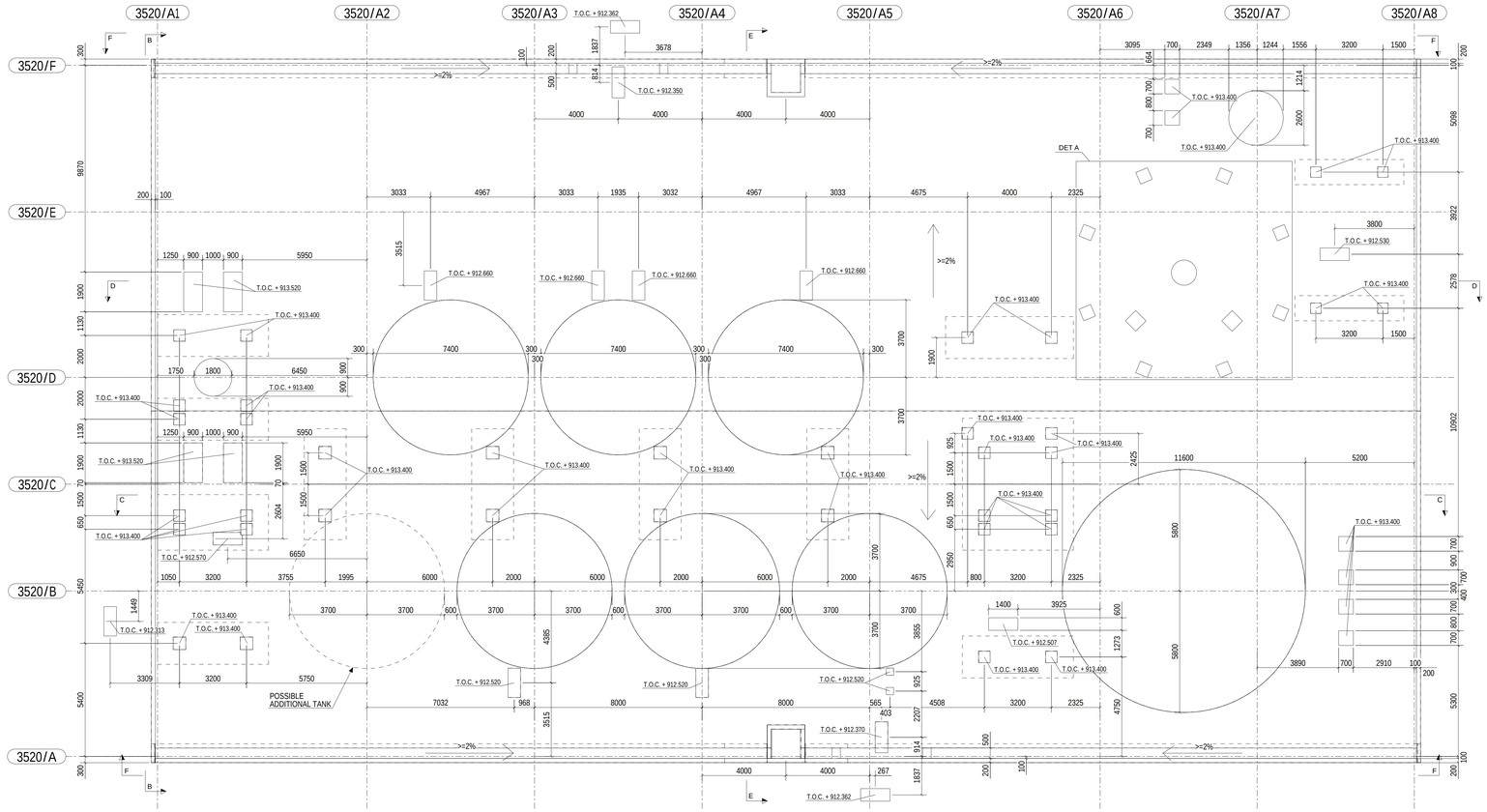
ATTACHMENT 2 OTE INPUT DRAWINGS(CIVIL WORK GUIDE DRAWING)

PROPRIETARY & CONFIDENTIAL

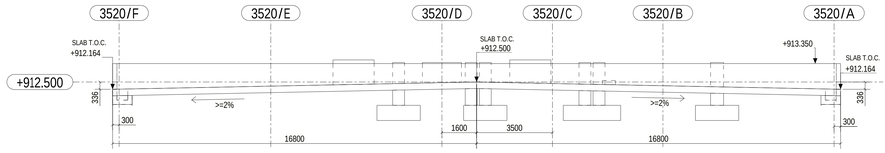
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DRAWING NUMBER: MD-216-3520-EG-CV-GAD-1001 | REV: A01

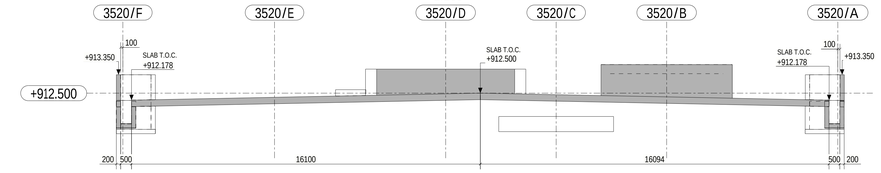
LEVEL +912.500
1:100



B - B
1:100



E - E
1:100



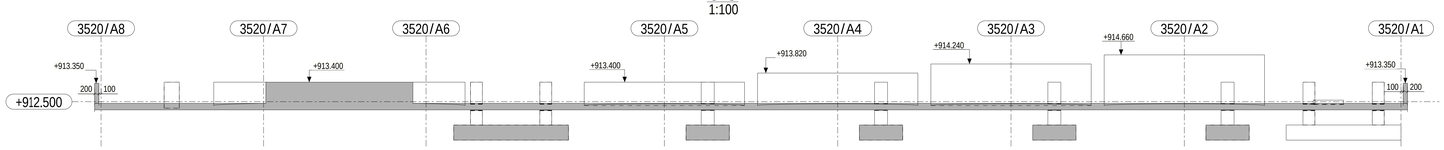
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3520 = RL(+)912.500 m

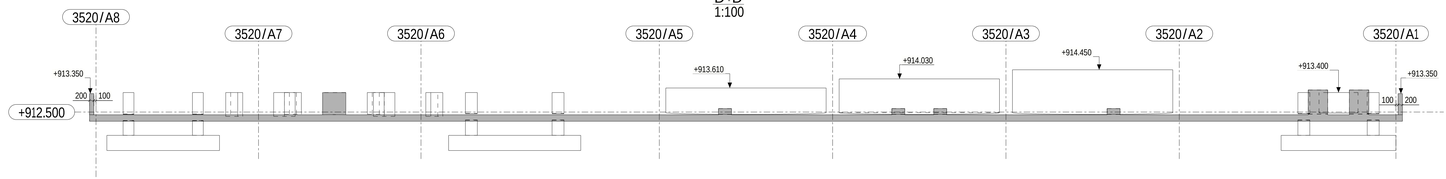
- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

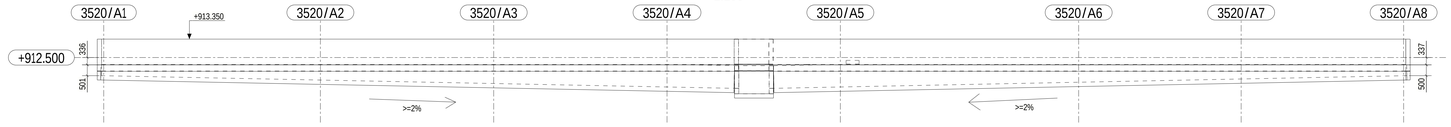
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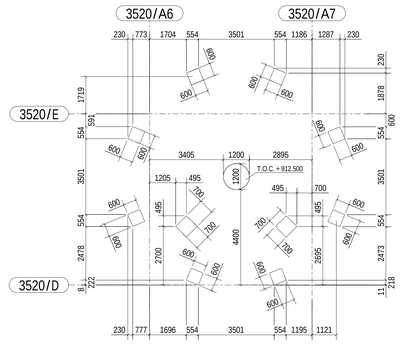
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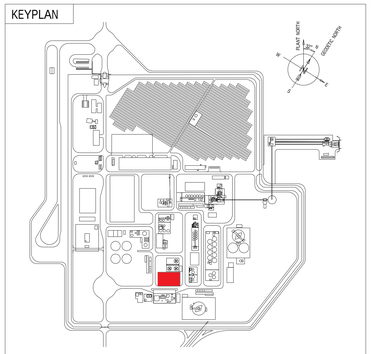
F - F
1:100



A - A
1:100
T.O.C. +913.400



KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3520-EG-ST-GAD-1001	3D VIEW	MD-216-3520-EG-ST-GAD-1002	PLAN AND SECTIONS
MD-216-3520-EG-ME-GAD-1043	SECTION D-D	MD-216-1000-EG-ST-CRT-0001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN BASIS
MD-216-3520-EG-ME-GAD-1041	SECTION A-A AND B-B	MD-216-3520-EG-ME-GAD-1042	SECTION C-C
MD-216-3520-EG-ME-GAD-1039	PLAN AT EL. +912.500	MD-216-3520-EG-ME-GAD-1040	PLAN VIEW

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	11.12.2019	ISSUED FOR APPROVAL	AmFor	MkLai	LuiPz	SuoCha	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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LARSEN & TOUBRO DRAWING NUMBER	019045_450_C_G_24943_001	REVISION	0
OUTOTEC DRAWING NUMBER	900707_GAU01_RN13_CTA04_35201	DOCUMENT ID	OU100667800
		REVISION	0

DRAWN	AmFor	11.12.2019
CHECKED	MkLai	11.12.2019
ENGINEER	LuiPz	11.12.2019
APPROVED	SuoCha	11.12.2019
EMPLOYER		
SCALE	1:100	A0

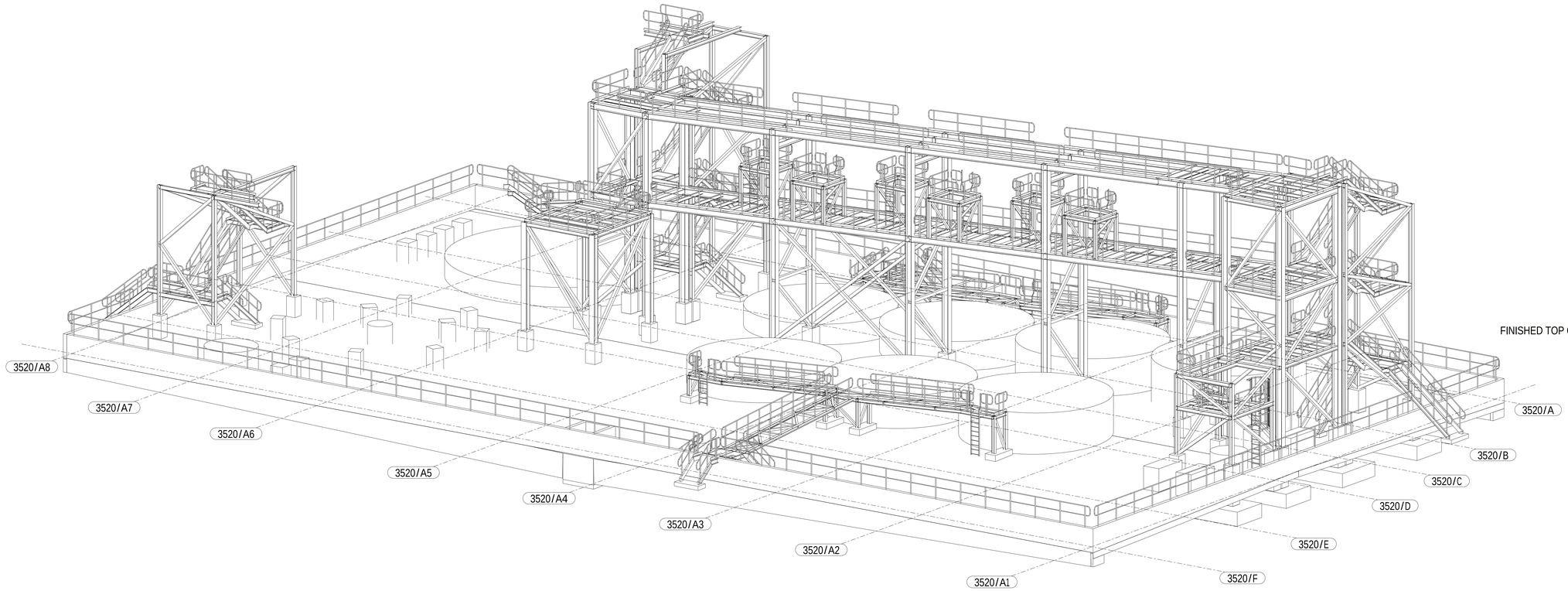
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
3520 NEUTRALIZATION AND THICKENING
PLAN AT EL. +912.500

DRAWING NUMBER	REV
MD-216-3520-EG-CV-GAD-1001	A01

Tekla Structures

DRAWING NUMBER: MD-216-3520-EG-ST-GAD-1001
REV: A01

NEUTRALIZATION AREA- 3D
1:80



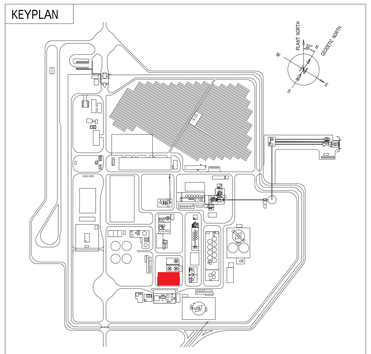
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3520 = RL(+)912.500 m

CHECKERED PLATE
GRATING
GRATING - FRP

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3520-EG-CV-GAD-1001	PLAN AT EL. +912.500	MD-216-3520-EG-ST-GAD-1002	PLAN AND SECTIONS
MD-216-3520-EG-ME-GAD-1043	SECTION D-D	MD-216-1000-EG-ST-CRT-0001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN BASIS
MD-216-3520-EG-ME-GAD-1041	SECTION A-A AND B-B	MD-216-3520-EG-ME-GAD-1042	SECTION C-C
MD-216-3520-EG-ME-GAD-1039	PLAN AT EL. +912.500	MD-216-3520-EG-ME-GAD-1040	PLAN VIEW

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	11.12.2019	ISSUED FOR APPROVAL	AmnFor	MkLai	LuiPiz	SuoCha	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER: O19045_450_S_B_28044_001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707_GAU01_RN13_CTA05_35201 — DOCUMENT ID: OU100667894 — REVISION: 0

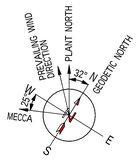
DRAWN	AmnFor	11.12.2019
CHECKED	MkLai	11.12.2019
ENGINEER	LuiPiz	11.12.2019
APPROVED	SuoCha	11.12.2019
EMPLOYER		
SCALE	1:100	A0

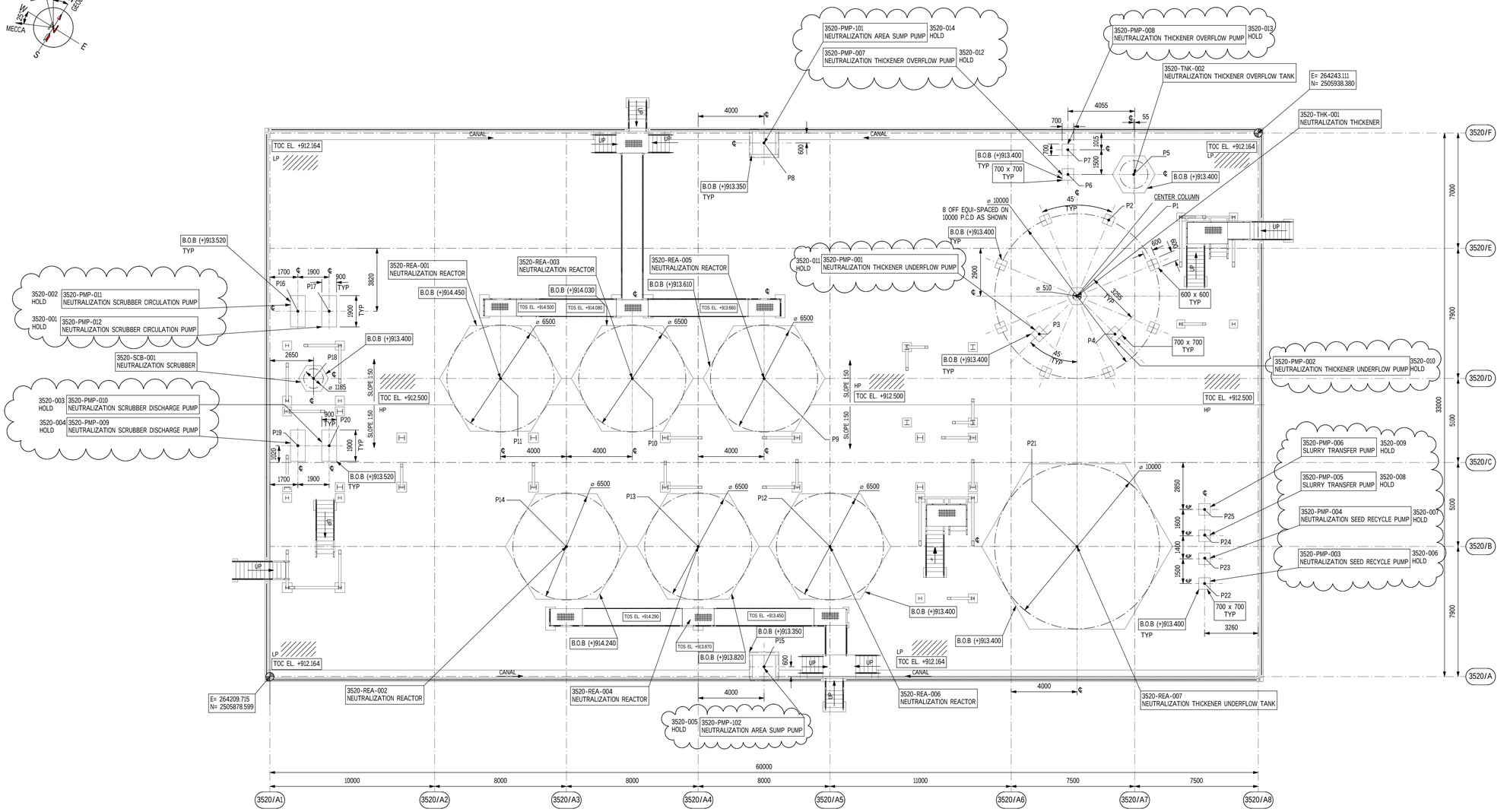
MANSOURAH - MASSARAH GOLD PROJECT
STEEL STRUCTURE GUIDE DRAWING
3520 NEUTRALIZATION AND THICKENING
3D VIEW

DRAWING NUMBER: MD-216-3520-EG-ST-GAD-1001 — REV: A01

Tekla Structures

DRAWING NUMBER: MD-216-3520-EG-ME-GAD-1008 | REV: A01





PLAN AT EL.(+) 912.500

* :-
FOR,
P1 & P2 : REFER DWG. MD-216-3520-EG-ME-GAD-1002 FOR LOAD DETAILS
P9, P10 & P11 : REFER DWG. MD-216-3520-EG-ME-PLN-1001 FOR LOAD DETAILS
P12, P13 & P14 : REFER DWG. MD-216-3520-EG-ME-PLN-1002 FOR LOAD DETAILS
P21 : REFER DWG. MD-216-3520-EG-ME-PLN-1007 FOR LOAD DETAILS

LOADING POINT	P1*	P2*	P3	P4	P5	P6	P7	P8	P9*	P10*	P11*	P12*	P13*	P14*	P15	P16	P17	P18	P19	P20	P21*	P22	P23	P24	P25
LOAD CASE	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]	Vz[kN] Hx[kN] Hy[kN]
1 DEAD LOAD EQUIPMENT			-6	-6	-30	-6	-6	-6							-6	-3	-3	-15	-2	-2		-4	-4	-5	-5
2 DEAD LOAD (SECONDARY EQUIPMENT)																									
3 OPERATION LOAD 1. FILLING																									
4 OPERATION LOAD 2. PRESSURE																									
5 OPERATION LOAD 3. TEMPERATURE																									
6 OPERATION LOAD 4. DYNAMIC			-9	-9		-12	-12	-12							-12	-6	-6		-5	-5		-10	-10	-10	-10
7 OPERATION LOAD 5. OTHERS																									
8 TEST LOAD FILLING / EMERGENCY																									
9 LIVE LOAD					-55													-15							
10 WIND ±X - DIRECTION																									
11 WIND ±Y - DIRECTION																									
12 EARTHQUAKE ±X - DIRECTION																									
13 EARTHQUAKE ±Y - DIRECTION																									
EQUIPMENT - NO.	3520-THK-001	3520-THK-001	3520-PMP-001	3520-PMP-002	3520-TNK-002	3520-PMP-007	3520-PMP-008	3520-PMP-101	3520-REA-005	3520-REA-003	3520-REA-001	3520-REA-006	3520-REA-004	3520-REA-002	3520-PMP-102	3520-PMP-011	3520-PMP-012	3520-SCB-001	3520-PMP-009	3520-PMP-010	3520-REA-007	3520-PMP-003	3520-PMP-004	3520-PMP-005	3520-PMP-006
DESCRIPTION	NEUTRALIZATION THICKNER	NEUTRALIZATION THICKNER	NEUTRALIZATION THICKNER UNDERFLOW PUMP	NEUTRALIZATION THICKNER UNDERFLOW PUMP	NEUTRALIZATION THICKNER OVERFLOW TANK	NEUTRALIZATION THICKNER OVERFLOW PUMP	NEUTRALIZATION THICKNER OVERFLOW PUMP	NEUTRALIZATION AREA SUMP PUMP	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION REACTOR	NEUTRALIZATION AREA SUMP PUMP	NEUTRALIZATION SCRUBBER CIRCULATION PUMP	NEUTRALIZATION SCRUBBER CIRCULATION PUMP	NEUTRALIZATION SCRUBBER	NEUTRALIZATION SCRUBBER DISCHARGE PUMP	NEUTRALIZATION SCRUBBER DISCHARGE PUMP	NEUTRALIZATION THICKNER UNDERFLOW TANK	NEUTRALIZATION SEED RECYCLE PUMP	NEUTRALIZATION SEED RECYCLE PUMP	SLURRY TRANSFER PUMP	SLURRY TRANSFER PUMP

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CHECKERED PLATE
GRATING
GRATING - FRP
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.
05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
LP = LOW POINT.
HP = HIGHT POINT.

DIRECTION OF FORCES AND MOMENTS: +Z, +Mz, +Y, +X, +My, +Mx

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT -WIND LOADS
- ADDITIONAL LOADS FOR ERECTION PURPOSES -EARTHQUAKE LOADS

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

GRATING 5 kN/m² | CONCRETE PLATFORMS 10 kN/m² | GROUNDFLOOR 25 kN/m² | CABLE FLOOR 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

DUST 0.5 kN/m² LIVE LOAD 0.75 kN/m²

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

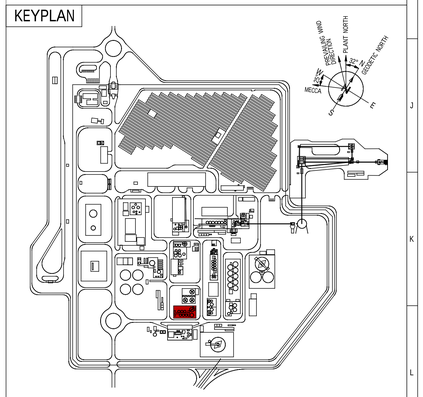
11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.500 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



0 1 10m

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3520-EG-ME-GAD-1039	GENERAL ARRANGEMENT DRAWING, PLAN AT EL +912.500
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3520-EG-ME-GAD-1009	PLAN AT EL +928.550

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	03-01-2020	ISSUED FOR APPROVAL	BS	TV	SC	RL	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

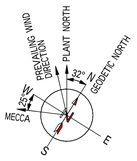
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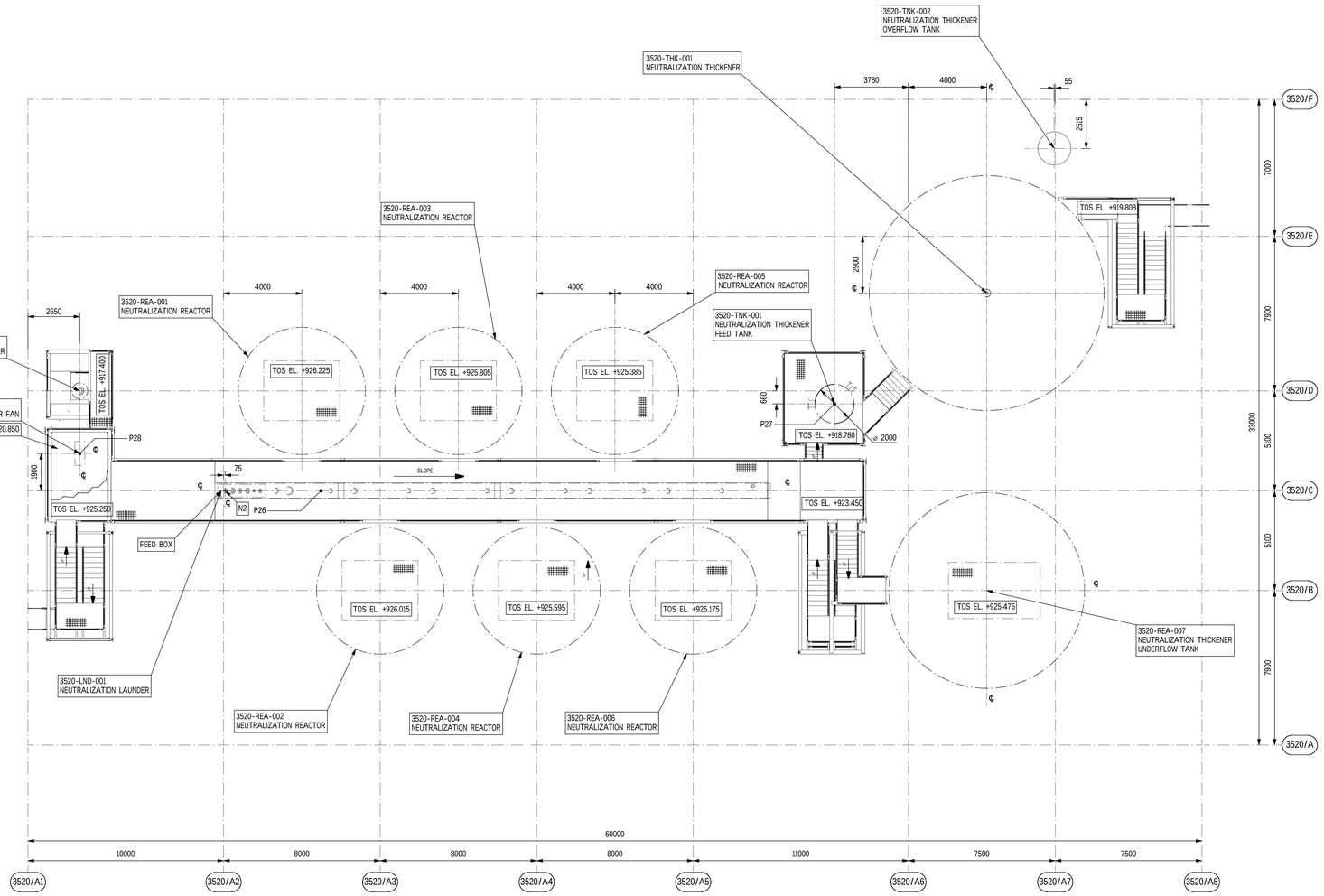
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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_450_M_D_02813_001			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707 _GAU01 _RN13 _MLD14 _35201		OU100667829	0
DRAWN	BS	03-01-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	03-01-2020	LOAD PLAN
ENGINEER	SC	03-01-2020	NEUTRALIZATION AND THICKENING
APPROVED	RL	03-01-2020	PLAN AT EL. +912.500
EMPLOYER			
SCALE 1:100	A0	DRAWING NUMBER MD-216-3520-EG-ME-GAD-1008	REV A01

DRAWING NUMBER: MD-216-3520-EG-ME-GAD-1009 | REV: A01





PLAN AT EL.(+)925.280

	CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
	CONCRETE FLOOR
A	CONCRETE FLOOR WITH ACID RESISTANT COATING
	CHECKERED PLATE
	GRATING
F	GRATING - FRP
	ACID RESISTANT BRICKLINING
	BRICK WALL
	CONCRETE WALL

GENERAL NOTES, LEGEND

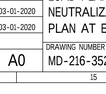
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LP = LOW POINT.
HP = HIGHT POINT.

DIRECTION OF FORCES AND MOMENTS



06 - THE LOADS SHOWN DO NOT INCLUDE :
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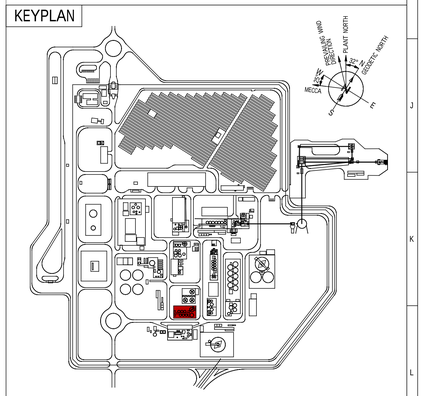
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* :-
FOR,
P26 : REFER DWG. MD-216-3520-EG-ME-GAD-1010 FOR LOAD DETAILS

	LOADING POINT	P26*				P27				P28			
	LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	My[kN]	Vz[kN]	Hx[kN]	Hy[kN]	My[kN]	Vz[kN]	Hx[kN]	Hy[kN]	
1	DEAD LOAD EQUIPMENT					- 21				- 30			
2	DEAD LOAD (SECONDARY EQUIPMENT)												
3	OPERATION LOAD 1. FILLING												
4	OPERATION LOAD 2. PRESSURE												
5	OPERATION LOAD 3. TEMPERATURE												
6	OPERATION LOAD 4. DYNAMIC									- 40			
7	OPERATION LOAD 5. OTHERS												
8	TEST LOAD FILLING / EMERGENCY												
9	LIVE LOAD					- 125							
10	WIND ±X - DIRECTION												
11	WIND ±Y - DIRECTION												
12	EARTHQUAKE ±X - DIRECTION												
13	EARTHQUAKE ±Y - DIRECTION												
	EQUIPMENT - NO.	3520-LND-001				3520-TNK-001				3520-FAN-001			
	DESCRIPTION	NEUTRALIZATION LAUNDER				NEUTRALIZATION THICKENER FEED TANK				NEUTRALIZATION SCRUBBER FAN			

KEYPLAN



0 1 10m

REV	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	03-01-2020 ISSUED FOR APPROVAL	BS	TV	SC	RL	

DOCUMENT NUMBER	REFERENCE DOCUMENTS
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MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3520-EG-ME-GAD-1008	PLAN AT EL. +912.150

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_M_D_02813_002 | REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ RN13 _ MLD14 _ 35202 | DOCUMENT ID: OU100667830 | REVISION: 0

DRAWN	BS	03-01-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	03-01-2020	LOAD PLAN
ENGINEER	SC	03-01-2020	NEUTRALIZATION AND THICKENING
APPROVED	RL	03-01-2020	PLAN AT EL. +925.280
EMPLOYER			DRAWING NUMBER: MD-216-3520-EG-ME-GAD-1009
SCALE 1:100	A0		REV: A01